

Exploratory Data Analysis for SPROJ

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PACKAGES

In order to do this Exploratory Data Analysis, I will use the following packages:

```
library(ggplot2) # for plotting data
library(readxl) # for reading excel files if necessary
library(dplyr) # for data wrangling
# library(arrow)
library(janitor) # for cleaning messy names of datasets
library(readr) # for reading files
library(gt) # for making 'publication ready' tables
library(usethis) #enables git pushing to put on github
library(gitcreds)
library(purrr) # this package is to loop and automate this entire process, instead of going t
library(fixest) #fixed fx
library(broom)
```

This senior project document will serve as the backbone of methodology and analysis part of the Senior Project”

Motivation

According to the UN Trade Hub and Development:

(<https://unctadstat.unctad.org/datacentre/reportInfo/US.RCA>)

“Revealed comparative advantage (RCA) is based on Ricardian trade theory, which posits that patterns of trade among countries are governed by their relative differences in productivity. Although such productivity differences are difficult to observe, an RCA metric can be readily calculated using trade data to”reveal” such differences. While the metric can be used to provide

a general indication and first approximation of a country's competitive export strengths, it should be noted that applied national measures which affect competitiveness such as tariffs, non-tariff measures, subsidies and others are not taken into account in the RCA metric."

"When a country has a revealed comparative advantage for a given product ($RCA > 1$), it is inferred to be a competitive producer and exporter of that product relative to a country producing and exporting that good at or below the world average. A country with a revealed comparative advantage in product i is considered to have an export strength in that product. The higher the value of a country's RCA for product i , the higher its export strength in product i ."

Part of this senior project aims to explain shifts in Revealed Comparative Advantage through a political economic lens, specifically in Mexico's Agriculture industry. As the UN mentions, this RCA is merely an inference variable, and is used as a 'first approximation' in comparative advantage. Many of the shifts in RCA over time are not captured. I will use classical economic theory, neoclassical theory, economic literature, and papers to describe shifts in the RCA through a political economic lens. My sector analysis in these documents will be done in order of sector code and number, starting from 01; Animals Live, throughout the rest of the agricultural sector.

The agricultural sector is one that is extremely broad- ranging from plants and livestock, to coffee beans and even honey. In order to further analyze the RCA index, we must discern what codes constitute the agriculture industry. Each sector has its own general product code at the 2 digit, 4 digit, and 6 digit level, with each digit being more specific in each product. Using the International Trade Administration (ITA) Harmonized System (HS) codes, we will use these definitions to discern what products are part of the agriculture industry. There is opinions in the literature that shows variations in the product codes that is part of the 'agricultural sector.' I will use the definitions from this paper (https://www.researchgate.net/publication/332551060_Changes_in_Foreign_Trade_in_Agricultural_Products) to classify the codes that belongs in the agricultural sector. Furthermore, I will analyze each sector at a general level, starting at the 2 digit, and then up to 4 digits. Due to constraints and limitations of the publicly available data, the maximum scope of analysis is up to the 4 digit HS code level. The data will also be split up into multiple time periods. Period A will be from 1992-2000, period B will be from 2000-2008, and period C will be from 2008-2016, period D will be from 2016-2024.

Functions

The section below describes the functions which are necessary in computing this analysis, I have made comments in the code chunks in order to further expand on each line of code and what it does.

Necessary Variables Function

```
## This is a function that takes in only the necessary variables for an RCA analysis, including  
  
necessary_variables <- function(df) { # inputs a dataframe  
  df |>  
    dplyr::mutate(cmdCode = as.character(cmdCode)) |>  
    arrange(period) |> #arranges the df by year  
    select(period, reporterISO, cmdCode, flowCode, qty, netWgt, fobvalue, primaryValue) #selects  
}
```

RCA Function

```
## RCA calculation function  
  
#inputs country exports of product i , country's total exports, world exports of product i, and world total  
calculate_rca <- function(country_product, country_total, world_product, world_total) {  
  
  #join by period to align years  
  rca_df <- country_product |>  
    inner_join(country_total, by = "period", suffix = c("_prod", "_total")) |>  
    inner_join(world_product, by = "period") |>  
    inner_join(world_total, by = "period", suffix = c("_world_prod", "_world_total")) |>  
    mutate(  
      RCA = (fobvalue_prod / fobvalue_total) /  
            (fobvalue_world_prod / fobvalue_world_total)  
    ) |>  
    select(period, RCA)  
  
  return(rca_df)  
}
```

generalized RCA Function

```
calculate_rca_generalized <- function(country_sector_df,  
                                     world_sector_df,  
                                     country_all_df,  
                                     world_all_df) {
```

```

#first, I aggregate by year and product, with variables period, and cmdCode, removing any NA
country_prod <- country_sector_df |>
  group_by(period, cmdCode) |>
  summarise(country_product_value = sum(fobvalue, na.rm = TRUE), .groups = "drop")

world_prod <- world_sector_df |>
  group_by(period, cmdCode) |>
  summarise(world_product_value = sum(fobvalue, na.rm = TRUE), .groups = "drop")

#next, I aggregate total exports using fobvalue (denominator)
country_all <- country_all_df |>
  group_by(period) |>
  summarise(total_country_value = sum(fobvalue, na.rm = TRUE), .groups = "drop")

world_all <- world_all_df |>
  group_by(period) |>
  summarise(total_world_value = sum(fobvalue, na.rm = TRUE), .groups = "drop")

# 3. next, I merge everything by year and commodity code
rca_df <- country_prod |>
  left_join(world_prod, by = c("period", "cmdCode")) |>
  left_join(country_all, by = "period") |>
  left_join(world_all, by = "period") |>
  mutate(
    RCA = (country_product_value / total_country_value) /
      (world_product_value / total_world_value),
    has_CA = RCA >= 1
  ) |>
  arrange(period, cmdCode)

return(rca_df) #outputs a dataframe
}

```

Necessary Dataframes

```

#mexico total exports time 1992 -> 2000

mx_xports_all <- read_csv("mx_total_xports_world/mx_xports_all.csv")

```

```
## world exports of 07 time 1992 -> 2000
```

```
world_xports_all <- read_csv("world_total_xports/world_xports_all.csv")
```

```
mx_xports_all <- necessary_variables(mx_xports_all) # dataframe for all of mexico's exports  
world_xports_all <- necessary_variables(world_xports_all) # dataframe for world exports of a
```

We need the dataframes above in order to complete the rest of the RCA Analysis. They are imperative in order so that the RCA variable can be calculated at a snapshot in time.

Sector Codes

```
## create sector codes
```

```
# in this instance, I will manually add the codes, as I will not be analyzing all sectors. T
```

```
sector_codes <- c("01", "02", "04", "05","07","08","09","10","11", "12", "13", "14", "15", "1
```

Process Sectors

```
process_sector <- function(sector) { ## sector in this instance is HS classification code
```

```
  mx_file <- paste0("mx_xports_prod_i/mx_xports_sector_", sector, ".csv") ## paste0 is used  
  world_file <- paste0("world_xports_prod_i/world_xports_sector_", sector, ".csv") ## simila
```

```
  # skip sector if files do not exist, if file does not exist in mx_file or file does not ex  
  if (!file.exists(mx_file) | !file.exists(world_file)) {  
    message("Skipping sector ", sector)  
    return(NULL)  
  }
```

```
  mx_sector <- read_csv(mx_file, show_col_types = FALSE) |> necessary_variables()  
  world_sector <- read_csv(world_file, show_col_types = FALSE) |> necessary_variables()
```

```
  rca_sector <- calculate_rca_generalized(  
    mx_sector,
```

```

    world_sector,
    mx_xports_all,
    world_xports_all
  )

  rca_sector$sector <- sector

  return(rca_sector)
}

```

Save all sectors in clean data folder

```

dir.create("data/clean/rca_sectors", recursive = TRUE, showWarnings = FALSE) #creates a directory

rca_big_df <- purrr::map_dfr(sector_codes, function(s) {
  result <- process_sector(s)
  if (!is.null(result)) { #if the result is not null
    write_csv(result, paste0("data/clean/rca_sectors/rca_sector_", s, ".csv")) # write the csv
    message("Saved sector ", s) #write message to save sector
  }
  return(result)
})

# add a sector column, and then add NAFTA dummy variable
rca_big_df <- rca_big_df |>
  mutate(
    cmdCode = as.character(cmdCode),
    sector = substr(cmdCode, 1, 2), # substr(string, start, stop), in this case start at character 1, stop at 2
    post_nafta = if_else(period >= 1994, 1, 0), # 1 for post nafta, 0 otherwise
    nafta_phase = case_when(
      period < 1994 ~ "pre", # period < 1994 make it pre
      period >= 1994 & period <= 2000 ~ "early", # period thru 1994 and 2000 titled early
      period > 2000 ~ "mature" #pweiod > 2000 titled after
    )
  )

```

Load Variables for Regression

```
# load exchange rate dataset and tidy up
exch_rate <- read_csv("regression/variables/exch_rate.csv")

exch_rate <- exch_rate |>
  select(Period, exch_rate) |>
  rename(period = "Period")

# load dataset for fdi inflow and tidy up

fdi_inflow <- read_csv("regression/variables/fdi_inflow.csv") # worldbank filtered out

fdi_inflow <- fdi_inflow |>
  select(Time, `Foreign direct investment, net inflows (BoP, current US$) [BX.KLT.DINV.CD.V]`) |>
  rename(period = "Time", fdiinflow = `Foreign direct investment, net inflows (BoP, current US$) [BX.KLT.DINV.CD.V]`) |>
  na.omit(fdi_inflow)

# load dataset for agriculture fdi

agriculture_fdi <- read_csv("regression/variables/agriculture_fdi.csv") ## https://www.gob.mx/agricultura/documentos/agricultura-fdi-2013-2018

agriculture_fdi <- agriculture_fdi |>
  select(period, agri_fdi)

rca_panel <- rca_big_df |>
  left_join(fdi_inflow, by = "period")|>
  left_join(exch_rate, by = "period") |>
  left_join(agriculture_fdi, by = "period")

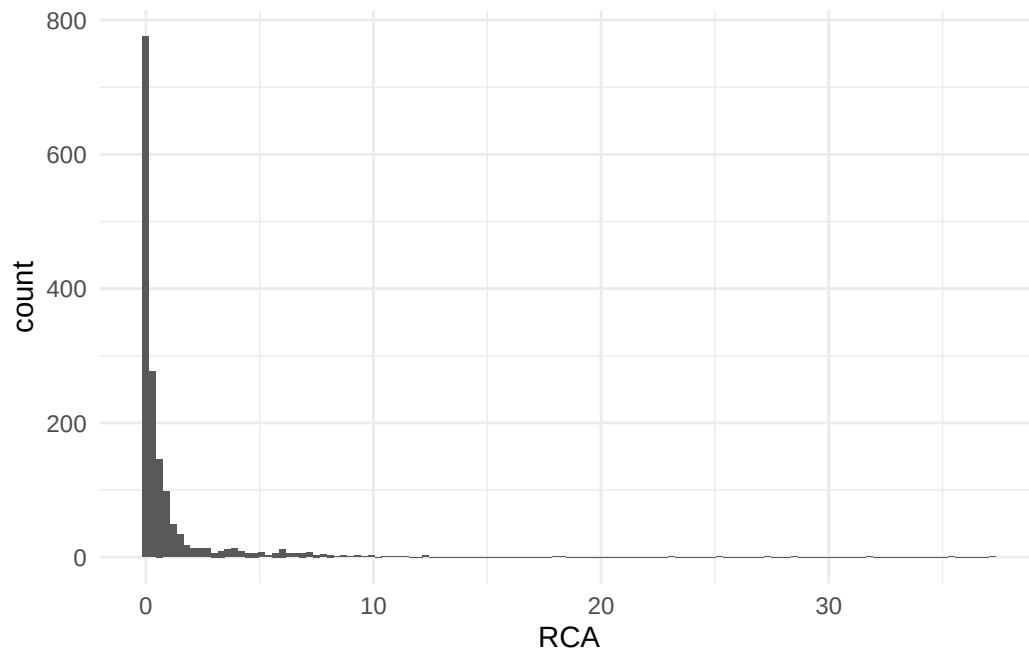
rca_panel <- rca_panel |>
  mutate(nafta_phase = factor(nafta_phase, levels = c("pre", "early", "mid", "late"))) # make
```

EDA

Now that we have our main dataframe `rca_panel`, we can conduct an EDA to see whether or not we need to do any transformations before we model.

1) Distribution

```
rca_panel |>
  select(RCA) |>
  ggplot(aes(x = RCA)) +
  geom_histogram(binwidth = 0.3) +
  theme_minimal()
```



We observe that our RCA is heavily right skewed. Hence, we *might* have to do a $\log(\text{RCA})$ transformation due to the nature of the distribution.

2) Time

Show the time frame (1992-2000)

```
rca_panel |>
  distinct(period) |>
  arrange(period)
```

```
# A tibble: 9 x 1
  period
  <dbl>
```

```
1 1992
2 1993
3 1994
4 1995
5 1996
6 1997
7 1998
8 1999
9 2000
```

3) Num of products

```
rca_panel |>
  distinct(cmdCode) |> #gives us distinct values for cmdCode
  nrow() #gives us the number of rows
```

```
[1] 191
```

The number of HSCodes that we are analyzing is 191.

4) Balanced

Does every product appear in every year?

```
rca_panel |>
  count(cmdCode) |>
  summary()
```

cmdCode	n
Length:191	Min. :1.000
Class :character	1st Qu.:9.000
Mode :character	Median :9.000
	Mean :8.262
	3rd Qu.:9.000
	Max. :9.000

5) Correlations

We must check for correlations between our explanatory variables

```
rca_panel |>
  distinct(period, post_nafta, exch_rate, fdiinflow, agri_fdi) |>
  select(post_nafta, exch_rate, agri_fdi) |>
  cor()
```

	post_nafta	exch_rate	agri_fdi
post_nafta	1.0000000	0.72444882	-0.44615789
exch_rate	0.7244488	1.00000000	-0.01091541
agri_fdi	-0.4461579	-0.01091541	1.00000000

6) Transform the variables

```
rca_panel <- rca_panel|>
  mutate(
    log_rca = log(RCA), # if RCA is right-skewed
    log_exch = log(exch_rate),
    asinh_fdi = asinh(agri_fdi) # sinh = (e^x - e^-x)/2
  )
```

MODELS

Model 1 Fixed Effects Regressiion

```
model <- feols(RCA ~ post_nafta + log(exch_rate) + log(fdiinflow) | cmdCode,
  data = rca_panel,
  cluster = ~cmdCode)

summary(model)
```

OLS estimation, Dep. Var.: RCA
Observations: 1,578
Fixed-effects: cmdCode: 191
Standard-errors: Clustered (cmdCode)

	Estimate	Std. Error	t value	Pr(> t)
post_nafta	0.117203	0.123747	0.947118	0.344782
log(exch_rate)	-0.199722	0.100374	-1.989787	0.048049 *
log(fdiinflow)	-0.110763	0.092438	-1.198246	0.232314

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

RMSE: 0.701706 Adj. R2: 0.920713

 Within R2: 0.019555

Model 2 FER

```
model2 <- feols(RCA ~ post_nafta + log(exch_rate) + asinh(agri_fdi) | cmdCode,
               data = rca_panel,
               cluster = ~cmdCode)
summary(model2)
```

OLS estimation, Dep. Var.: RCA

Observations: 1,578

Fixed-effects: cmdCode: 191

Standard-errors: Clustered (cmdCode)

	Estimate	Std. Error	t value	Pr(> t)
post_nafta	-0.018465	0.064710	-0.285348	0.775688
log(exch_rate)	-0.203169	0.109330	-1.858319	0.064671 .
asinh(agri_fdi)	-0.034516	0.019406	-1.778586	0.076907 .

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

RMSE: 0.701757 Adj. R2: 0.920702

 Within R2: 0.019413

Model 3

```
model3 <- feols(log_rca ~ post_nafta + log_exch + asinh_fdi | cmdCode,
               data = rca_panel,
               cluster = ~cmdCode)
summary(model3)
```

OLS estimation, Dep. Var.: log_rca

Observations: 1,578

```

Fixed-effects: cmdCode: 191
Standard-errors: Clustered (cmdCode)
              Estimate Std. Error  t value   Pr(>|t|)
post_nafta -0.412730    0.151063 -2.73217 0.00688526 **
log_exch    0.429726    0.128520  3.34366 0.00099595 ***
asinh_fdi   -0.142485    0.051818 -2.74971 0.00654076 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
RMSE: 1.17519      Adj. R2: 0.827889
                Within R2: 0.009527

```

Model 4 USE THIS

```

model4 <- feols(log_rca ~ nafta_phase + log(exch_rate) + asinh(agri_fdi)|cmdCode,
                data = rca_panel,
                cluster = ~cmdCode)

summary(model4)

```

```

OLS estimation, Dep. Var.: log_rca
Observations: 1,578
Fixed-effects: cmdCode: 191
Standard-errors: Clustered (cmdCode)
              Estimate Std. Error  t value   Pr(>|t|)
nafta_phaseearly -0.412730    0.151063 -2.73217 0.00688526 **
log(exch_rate)    0.429726    0.128520  3.34366 0.00099595 ***
asinh(agri_fdi)   -0.142485    0.051818 -2.74971 0.00654076 **
... 2 variables were removed because of collinearity (nafta_phasemid and nafta_phaselate)
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
RMSE: 1.17519      Adj. R2: 0.827889
                Within R2: 0.009527

```

Interpretation

The interpretation of model 4 will be as follows.

Log(Exchange Rate)

A 1 percent increase in the exchange rate is associated with a 0.43 percent increase in RCA (at the 1 percent level)

nafta__phaseearly

Log(RCA) was 0.41 (units) lower than in the pre-NAFTA baseline, holding exchange rate and agriculture_FDI constant (significant at the 5 percent level)

**** asinh(agri_fdi)****

A rise in agriculture FDI is associated with a decrease in RCA (counterintuitive).

A 1 percent increase in FDI in Mexico's Agriculture Sector is linked with a -0.142 percent decrease in RCA, holding all else constant.

- FDI may have been used to increase domestic production rather than building export competitiveness in the global market.
- FDI may flow into sectors that lose comparative advantage.
- Aggregate agriculture FDI may have been flowing disproportionately into subsectors (i.e fertilizer, etc..) so FDI is not actually reaching the products driving the RCA variation.

TABLES

```
# Main Model Table (Claude AI)

model4_tidy <- broom::tidy(model4, conf.int = TRUE) |>
  mutate(
    term = case_when(
      term == "nafta_phaseearly" ~ "NAFTA Phase: Early",
      term == "nafta_phasepost" ~ "NAFTA Phase: Post",
      term == "log(exch_rate)" ~ "Log Exchange Rate",
      term == "asinh(agri_fdi)" ~ "IHS Agricultural FDI",
      TRUE ~ term
    ),
    stars = case_when(
      p.value < 0.01 ~ "***",
      p.value < 0.05 ~ "**",
      p.value < 0.1 ~ "*",
      TRUE ~ ""
    ),
    estimate_stars = paste0(round(estimate, 3), stars),
    std.error = paste0("(", round(std.error, 3), ")")
  ) |>
  select(term, estimate_stars, std.error, p.value)
```

```

# Build gt table
model4_tidy |>
  gt() |>
  tab_header(
    title = "Fixed Effects Estimation of NAFTA's Impact on RCA",
    subtitle = "Dependent variable: Log(RCA)"
  ) |>
  cols_label(
    term = "Variable",
    estimate_stars = "Coefficient",
    std.error = "Std. Error",
    p.value = "p-value"
  ) |>
  fmt_number(
    columns = p.value,
    decimals = 3
  ) |>
  tab_spanner(
    label = "Log(RCA)",
    columns = c(estimate_stars, std.error, p.value)
  ) |>
  tab_footnote(
    footnote = "*** p < 0.01, ** p < 0.05, * p < 0.1" #shows statistical significance
  ) |>
  tab_footnote(
    footnote = "Clustered standard errors at the cmdCode level. Reference category: Pre-NAFTA"
  ) |>
  tab_source_note(
    source_note = "Source: UN Comtrade, Secretaría de Economía, Banco de México."
  ) |>
  tab_options(
    heading.title.font.size = 14,
    heading.subtitle.font.size = 12,
    column_labels.font.weight = "bold",
    table.width = pct(100)
  )

```

Fixed Effects Estimation of NAFTA's Impact on RCA

Dependent variable: Log(RCA)] Fixed Effects Estimation of NAFTA's Impact on RCA

Variable	Log(RCA)	
	Coefficient	Std. Error
NAFTA Phase: Early	-0.413***	(0.151)
Log Exchange Rate	0.43***	(0.129)
IHS Agricultural FDI	-0.142***	(0.052)

*** p < 0.01, ** p < 0.05, * p < 0.1

Clustered standard errors at the cmdCode level. Reference category: Pre-NAFTA.

Source: UN Comtrade, Secretaría de Economía, Banco de México.

SECTORS

01 Animals; Live

This section will calculate RCA over time before NAFTA of the animal section, containing HS product code of 0101 to 0106.

```
## load the datasets
mx_xports_sector_01 <- read_csv("mx_xports_prod_i/mx_xports_sector_01.csv")
world_xports_sector_01 <- read_csv("world_xports_prod_i/world_xports_sector_01.csv")

## necessary variables

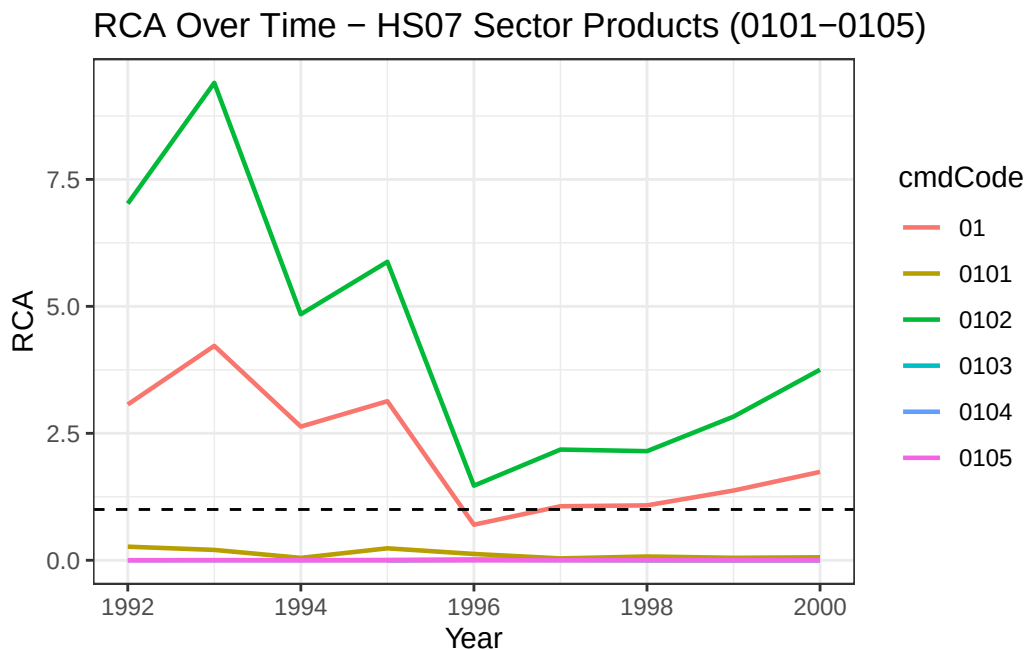
mx_xports_sector_01 <- necessary_variables(mx_xports_sector_01)
world_xports_sector_01 <- necessary_variables(world_xports_sector_01)

# calculate RCA

rca_sector_01 <- calculate_rca_generalized(mx_xports_sector_01, world_xports_sector_01, mx_xports_sector_01)
```

ggplot sector 01 from 1992 - 2000.

```
ggplot(rca_sector_01, aes(x = period, y = RCA, color = cmdCode, group = cmdCode)) +
  geom_line(size = 0.8) +
  geom_hline(yintercept = 1, linetype = "dashed") +
  labs(title = "RCA Over Time - HS07 Sector Products (0101-0105)",
       x = "Year", y = "RCA") +
  theme_bw()
```



Interesting that Mexico seems to have a general RCA of Live Animals, but there was a sharp decline starting in 1993. NAFTA was enacted January 1, 1994- was it a driving factor in RCA shift? What about data from future years?

02 Meat and Edible Meat offal

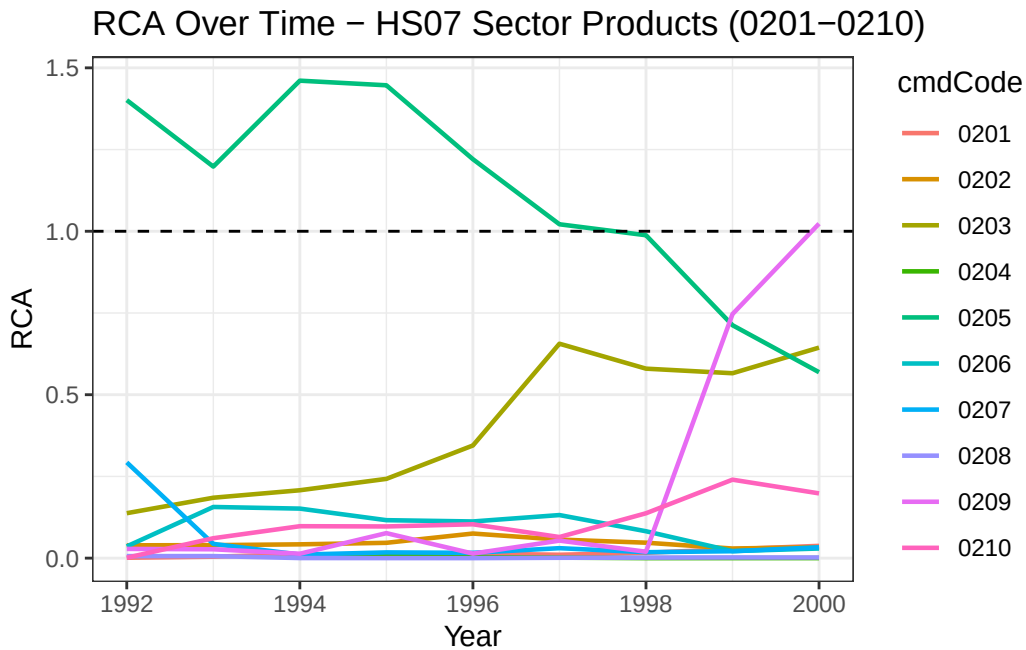
```
## load the dataframes
mx_xports_sector_02 <- read_csv("mx_xports_prod_i/mx_xports_sector_02.csv")
world_xports_sector_02 <- read_csv("world_xports_prod_i/world_xports_sector_02.csv")

# overwrite with necessary variables

mx_xports_sector_02 <- necessary_variables(mx_xports_sector_02)
world_xports_sector_02 <- necessary_variables(world_xports_sector_02)
```

```
## calculate rca
rca_sector_02 <- calculate_rca_generalized(mx_xports_sector_02, world_xports_sector_02, mx_xports_sector_02)

## graph RCA
ggplot(rca_sector_02, aes(x = period, y = RCA, color = cmdCode, group = cmdCode)) +
  geom_line(size = 0.8) +
  geom_hline(yintercept = 1, linetype = "dashed") +
  labs(title = "RCA Over Time - HS07 Sector Products (0201-0210)",
       x = "Year", y = "RCA") +
  theme_bw()
```



Sector of 02 seems to not have much of a comparative advantage in Mexico.

04 Dairy Produce; Birds' eggs; Natural Honey; Edible products of animal origin

04, 0401 -> 0410 EXCLUDING HONEY 0409

First, we load the dataframes necessary, and overwrite them with necessary variables

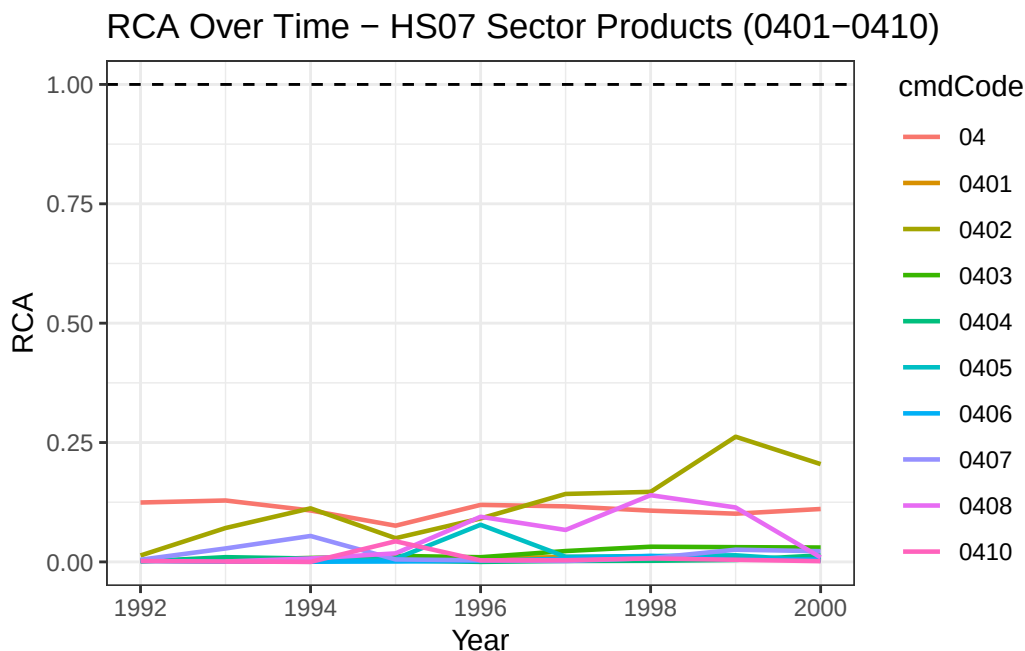
```
mx_xports_sector_04 <- read_csv("mx_xports_prod_i/mx_xports_sector_04.csv")
world_xports_sector_04 <- read_csv("world_xports_prod_i/world_xports_sector_04.csv")
```

```
mx_xports_sector_04 <- necessary_variables(mx_xports_sector_04)
world_xports_sector_04 <- necessary_variables(world_xports_sector_04)
```

Calculate RCA and then graph

```
rca_sector_04 <- calculate_rca_generalized(mx_xports_sector_04, world_xports_sector_04, mx_xports_sector_04)
```

```
ggplot(rca_sector_04, aes(x = period, y = RCA, color = cmdCode, group = cmdCode)) +
  geom_line(size = 0.8) +
  geom_hline(yintercept = 1, linetype = "dashed") +
  labs(title = "RCA Over Time - HS07 Sector Products (0401-0410)",
       x = "Year", y = "RCA") +
  theme_bw()
```



05 Other Animal Products

ANIMAL ORIGINATED PRODUCTS; NOT ELSEWHERE SPECIFIED OR INCLUDED

In this sector, I will avoid

- 0501 Human hair; unworked, whether or not washed or scoured; waste of human hair
- 0509 Sponges

- 0511 Animal products not elsewhere specified or included; dead animals of chapter 1 or 3, unfit for human consumption

```
mx_xports_sector_05 <- read_csv("mx_xports_prod_i/mx_xports_sector_05.csv")

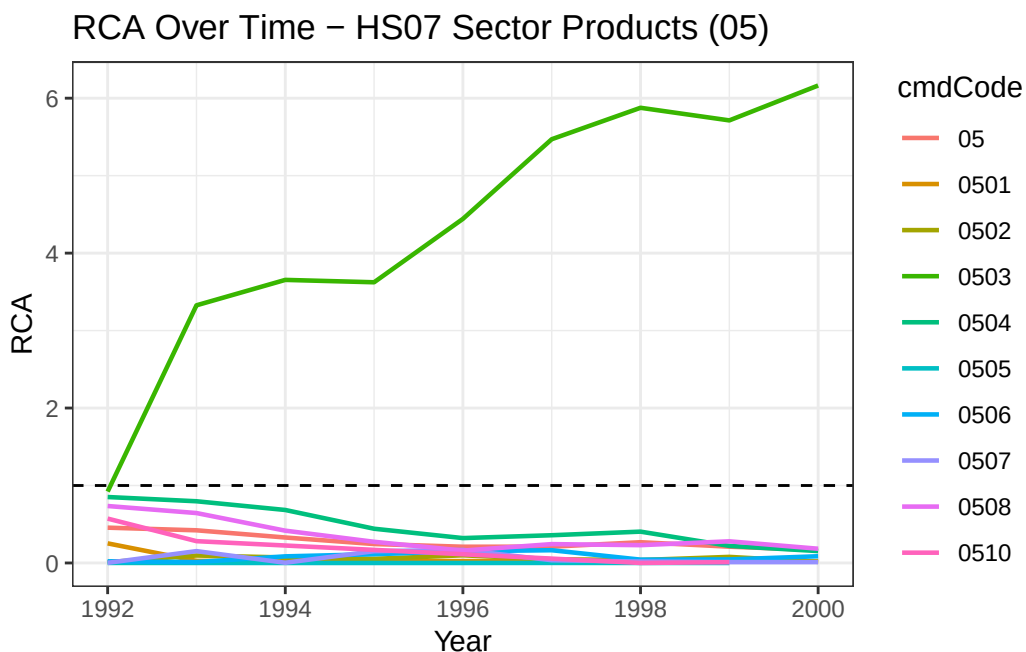
world_xports_sector_05 <- read_csv("world_xports_prod_i/world_xports_sector_05.csv")

mx_xports_sector_05 <- necessary_variables(mx_xports_sector_05)
world_xports_sector_05 <- necessary_variables(world_xports_sector_05)
```

Calculate RCA and then graph

```
rca_sector_05 <- calculate_rca_generalized(mx_xports_sector_05, world_xports_sector_05, mx_xports_sector_05)

ggplot(rca_sector_05, aes(x = period, y = RCA, color = cmdCode, group = cmdCode)) +
  geom_line(size = 0.8) +
  geom_hline(yintercept = 1, linetype = "dashed") +
  labs(title = "RCA Over Time - HS07 Sector Products (05)",
       x = "Year", y = "RCA") +
  theme_bw()
```



Interestingly, 0503 “Horsehair, waste” has a comparative advantage in that product, but the sector does not have a comparative advantage.

07- Vegetables and Certain Roots and Tubes; Edible

```
# read the datasets in each folder

#mexico exports of 07 VEGETABLES AND CERTAIN ROOTS AND TUBERS; EDIBLE time 1992 -> 2000
mx_xports_07 <- read_csv("mx_xports_prod_i/mx_xports_07.csv")

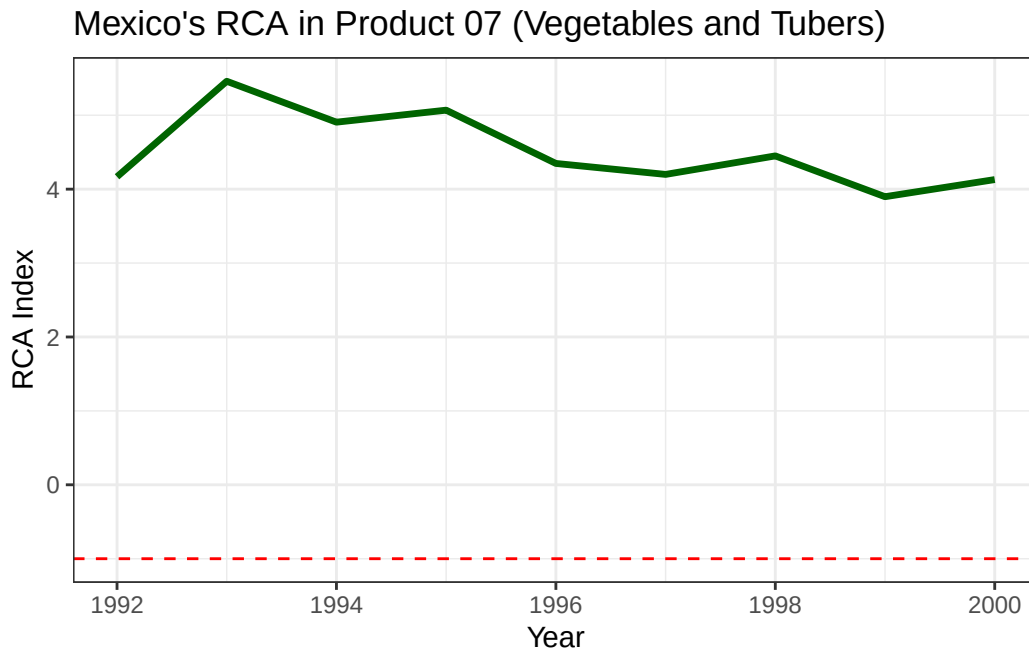
world_xports_07 <- read_csv("world_xports_prod_i/world_xports_07.csv")


mx_xports_07 <- necessary_variables(mx_xports_07)
mx_xports_all <- necessary_variables(mx_xports_all)
world_xports_07 <- necessary_variables(world_xports_07)
world_xports_all <- necessary_variables(world_xports_all)


## apply to dataframes

mx_rca_07_1992_2000 <- calculate_rca(
  mx_xports_07,
  mx_xports_all,
  world_xports_07,
  world_xports_all)


ggplot(mx_rca_07_1992_2000, aes(x = period, y = RCA)) +
  geom_line(color = "darkgreen", size = 1.2) +
  geom_hline(yintercept = -1, linetype = "dashed", color = "red") +
  labs(
    title = "Mexico's RCA in Product 07 (Vegetables and Tubers)",
    x = "Year",
    y = "RCA Index"
  ) +
  theme_bw()
```



Sector Analysis : 07

Now, I am tasked to clean the dataset and calculate an RCA given the dataset with the entire sector. Using the code above, find a way to generalize the RCA calculation to continue the calculation, with an ultimate goal of showing an RCA graph for each product in the sector.

```
# read the dataframe of the 07 sector containing country MX exports of product 0701, 0702, 0703, 0704, 0705, 0706, 0707, 0708, 0709, 0710
#mexico exports of products 0701 to 0710 and time 1992 -> 2000

mx_xports_sector_07 <- read_csv("mx_xports_prod_i/mx_xports_sector_07.csv")
```

Now, I must use the necessary variables for the RCA. necessary variables are the following: period, reporterISO, cmdCode, flowCode, qty, netWgt, fobvalue, primaryValue. Recall that the RCA variable is calculated with the following:

$RCA = [(Country\ Exports\ of\ Product\ i / Country\ A\ Exports\ of\ ALL\ products) / (World\ Exports\ of\ product\ i / World\ Exports\ of\ all\ products)]$

$$RCA_i = \frac{\frac{X_{A,i}}{X_{A,all}}}{\frac{X_{World,i}}{X_{World,all}}}$$

The RCA variable calculates a share of country exports a product over the world exports of said product. If the RCA variable is greater than, or equal to 1, then the country has a revealed

comparative advantage in producing said good. According to neoclassical theory, gains from trade are still probable in comparative advantage. Theory says to export goods in which the country has a comparative advantage in producing.

```
mx_xports_sector_07 <- necessary_variables(mx_xports_sector_07)
```

Now, I have the necessary variables, Country MX Exports of product 0701 -> 0710. I have the data of Mexico's exports of all products, titled `mx_xports_all`. Now, I need World's exports of products 0701 -> 0710. I can do this by using UNCOMTRADE data as Ill. Repeating the same steps, I will grab the necessary variables.

```
#load the dataframe of world exports of prod 0701 -> 0710
world_xports_sector_07 <- read_csv("world_xports_prod_i/world_xports_sector_07.csv")

#grab necessary variables.
world_xports_sector_07 <- necessary_variables(world_xports_sector_07)
```

I now have World exports of product 0701 -> 0710 with the same time period of 1992 to 2000. Recall I have datasets of Mexico's exports of all products titled `mx_xports_all` with variables period, reporterISO, cmdCode, flowCode, qty, netWgt, fobvalue, primaryValue. I also have `world_xports_all` which contains world exports of all products with variables period, reporterISO, cmdCode, flowCode, qty, netWgt, fobvalue, primaryValue. However, notice that `world_xports_sector_07` and `mx_xports_sector_07` have the cmdCode, however, they are not aggregated, and their cmdCode varies from 0701 -> 0710. Below is an attempted generalized calculation of RCA, with a clear difference of grouping by `group_by(period, cmdCode)` due to the nature of the dataframe with varying cmdCode

Let's test this in action to see if it works!

RCA of sector 0701 to 0710.

```
# inputs need: country_sector_df, world_sector_df, country_all_df, world_all_df

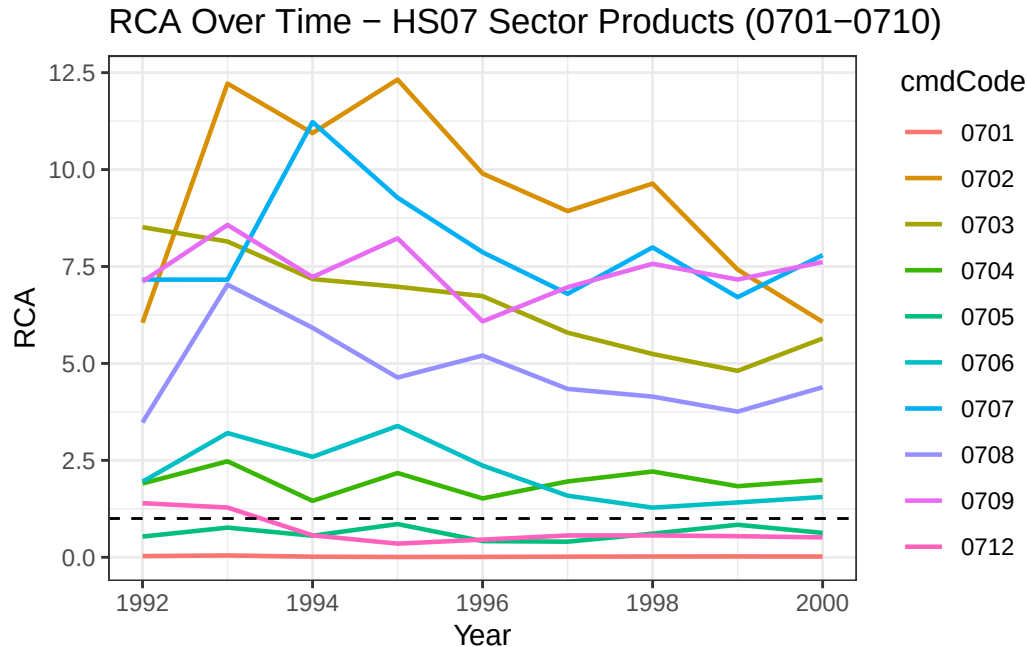
# have: mx_xports_sector_07, world_xports_sector_07, mx_xports_all, world_xports_all

rca_sector_07 <- calculate_rca_generalized(mx_xports_sector_07, world_xports_sector_07, mx_xp
```

Visualize over time:

```
## ggplot to bisualize over time
ggplot(rca_sector_07, aes(x = period, y = RCA, color = cmdCode, group = cmdCode)) +
  geom_line(size = 0.8) +
```

```
geom_hline(yintercept = 1, linetype = "dashed") +
labs(title = "RCA Over Time - HS07 Sector Products (0701-0710)",
     x = "Year", y = "RCA") +
theme_bw()
```



I can now develop code to show which products have an $RCA > 1$ and $RCA < 1$.

I can do this by running a simple filter:

```
## show which years had an RCA > 1
table_RCA_gt1 <- rca_sector_07 |>
  filter(RCA > 1) |>
  arrange(period, cmdCode)
table_RCA_gt1
```

A tibble: 65 x 8

	period	cmdCode	country_product_value	world_product_value	total_country_value
	<dbl>	<chr>	<dbl>	<dbl>	<dbl>
1	1992	0702	166984992	1467571657	184779539480
2	1992	0703	123081000	768524226	184779539480
3	1992	0704	14027000	391613956	184779539480
4	1992	0706	10529000	287038881	184779539480
5	1992	0707	91608000	679846342	184779539480

```

6   1992 0708                16944000                258995555                184779539480
7   1992 0709                302278016                2260100293                184779539480
8   1992 0712                17868000                 680654076                184779539480
9   1993 0702                394972000                1762452605                207545660288
10  1993 0703                146144000                978346042                 207545660288
# i 55 more rows
# i 3 more variables: total_world_value <dbl>, RCA <dbl>, has_CA <lgl>

```

or by showing which years had an RCA > 1

```

#summarize the comparative advantage by product by grouping by product code
summary_CA_by_product <- rca_sector_07 |>
  group_by(cmdCode) |>
  summarise(
    years_with_CA = paste(period[RCA > 1], collapse = ", "),
    n_years_CA = sum(RCA > 1),
    total_years = n(),
    share_years_CA = n_years_CA / total_years
  )

```

and a publication ready table:

```

summary_CA_by_product |>
  gt() |>
  tab_header(
    title = "Mexico's RCA, Sector 07 (RCA > 1)",
  )

```

I can fix this dataframe by changing the name of the previous one

Sector definitions

07 - VEGETABLES AND CERTAIN ROOTS AND TUBERS; EDIBLE

- 0701 - Potatoes; fresh or chilled
- 0702 - Tomatoes; fresh or chilled
- 0703 - Onions, shallots, garlic, leeks and other alliaceous vegetables; fresh or chilled
- 0704 - Cabbages, caulifloIrs, kohlrabi, kale and similar edible brassicas; fresh or chilled
- 0705 - Lettuce (lactuca sativa) and chicory (cichorium spp.) fresh or chilled
- 0706 - Carrots, turnips, salad beetroot, salsify, celeriac, radishes and similar edible roots; fresh or chilled

[

Mexico's RCA, Sector 07 (RCA > 1)] Mexico's RCA, Sector 07 (RCA > 1)

cmdCode	years_with_CA	n_years_CA	total_years	share_y
0701		0	9	
0702	1992, 1993, 1994, 1995, 1996, 1997, 1998, 1999, 2000	9	9	
0703	1992, 1993, 1994, 1995, 1996, 1997, 1998, 1999, 2000	9	9	
0704	1992, 1993, 1994, 1995, 1996, 1997, 1998, 1999, 2000	9	9	
0705		0	9	
0706	1992, 1993, 1994, 1995, 1996, 1997, 1998, 1999, 2000	9	9	
0707	1992, 1993, 1994, 1995, 1996, 1997, 1998, 1999, 2000	9	9	
0708	1992, 1993, 1994, 1995, 1996, 1997, 1998, 1999, 2000	9	9	
0709	1992, 1993, 1994, 1995, 1996, 1997, 1998, 1999, 2000	9	9	
0712	1992, 1993	2	9	

- 0707 - Cucumbers and gherkins; fresh or chilled
- 0708 - Leguminous vegetables; shelled or unshelled, fresh or chilled
- 0709 - Vegetables; n.e.c. in chapter 07, fresh or chilled
- 0710 - Vegetables (uncooked or cooked by steaming or boiling in water); frozen
- 0711 - Vegetables provisionally preserved; (e.g. by sulphur dioxide gas, in brine, in sulphur water or in other preservative solutions), but unsuitable in that state for immediate consumption
- 0712 - Vegetables, dried; whole, cut, sliced, broken or in powder, but not further prepared
- 0713 - Vegetables, leguminous; shelled, whether or not skinned or split, dried

08- FRUIT AND NUTS, EDIBLE; PEEL OF CITRUS FRUIT OR MELONS

We omit the following:

- 0812 Fruit and nuts provisionally preserved; e.g. by sulphur dioxide gas, brine, in sulphur water or in other preservative solutions, but unsuitable in that state for immediate consumption
- 0813 Fruit, dried, other than that of heading no. 0801 to 0806; mixtures of nuts or dried fruits of this chapter
- 0814 Peel of citrus fruit or melons (including watermelons); fresh, frozen dried or provisionally preserved in brine, in sulphur water or in other preservative solutions

```
mx_xports_sector_08 <- read_csv("mx_xports_prod_i/mx_xports_sector_08.csv")

world_xports_sector_08 <- read_csv("world_xports_prod_i/world_xports_sector_08.csv")

mx_xports_sector_05 <- necessary_variables(mx_xports_sector_08)
world_xports_sector_05 <- necessary_variables(world_xports_sector_08)
```

RCA of sector 0801 to 0810.

```
# inputs need: country_sector_df, world_sector_df, country_all_df, world_all_df

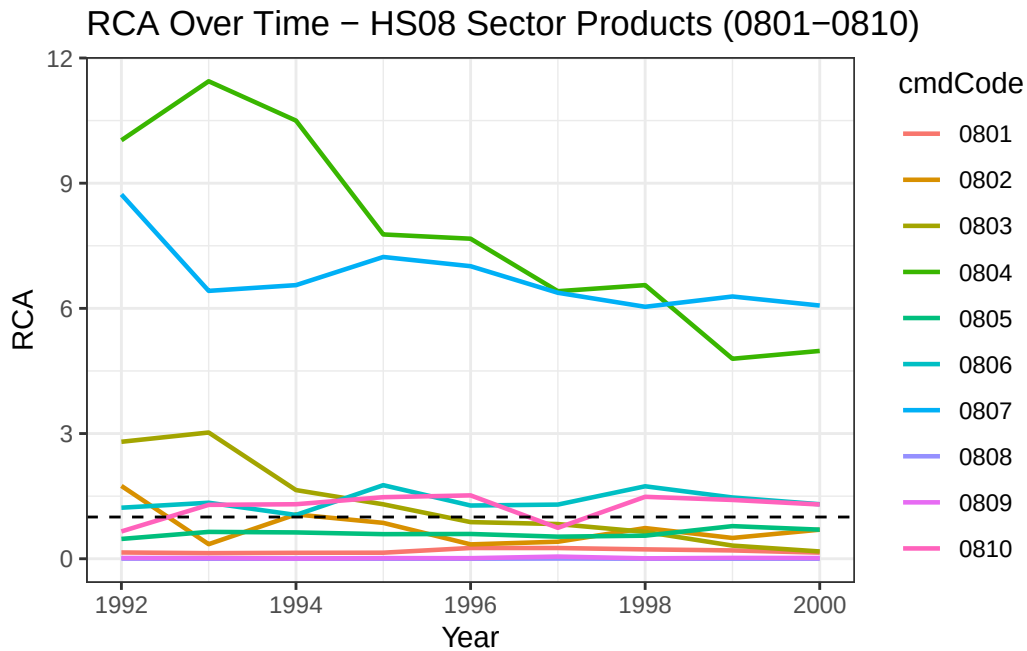
# have: mx_xports_sector_07, world_xports_sector_07, mx_xports_all, world_xports_all

rca_sector_08 <- calculate_rca_generalized(mx_xports_sector_08, world_xports_sector_08, mx_xp
```

Plot sector 08

Visualize over time:

```
## ggplot to bisualize over time
ggplot(rca_sector_08, aes(x = period, y = RCA, color = cmdCode, group = cmdCode)) +
  geom_line(size = 0.8) +
  geom_hline(yintercept = 1, linetype = "dashed") +
  labs(title = "RCA Over Time - HS08 Sector Products (0801-0810)",
       x = "Year", y = "RCA") +
  theme_bw()
```



09- COFFEE, TEA, MATE AND SPICES

No 4 digit HS classification code is omitted.

```
mx_xports_sector_09 <- read_csv("mx_xports_prod_i/mx_xports_sector_09.csv")
world_xports_sector_09 <- read_csv("world_xports_prod_i/world_xports_sector_09.csv")

mx_xports_sector_09 <- necessary_variables(mx_xports_sector_09)
world_xports_sector_09 <- necessary_variables(world_xports_sector_09)
```

RCA of sector 0901 to 0910.

```
# inputs need: country_sector_df, world_sector_df, country_all_df, world_all_df

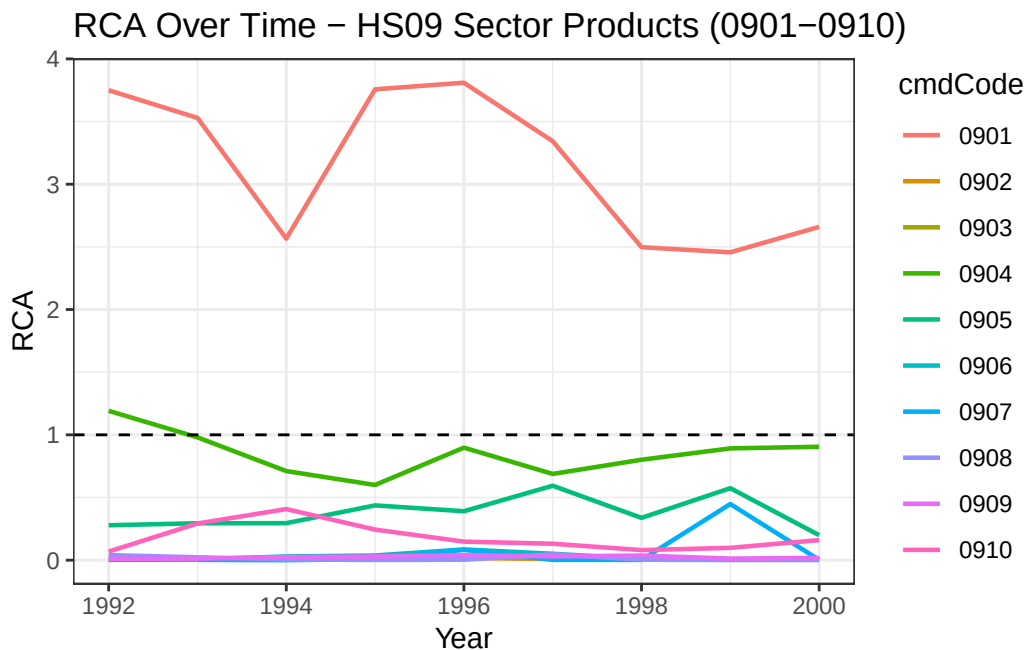
# have: mx_xports_sector_07, world_xports_sector_07, mx_xports_all, world_xports_all

rca_sector_09 <- calculate_rca_generalized(mx_xports_sector_09, world_xports_sector_09, mx_xp
```

Plot sector 09

Visualize over time:

```
## ggplot to bisualize over time
ggplot(rca_sector_09, aes(x = period, y = RCA, color = cmdCode, group = cmdCode)) +
  geom_line(size = 0.8) +
  geom_hline(yintercept = 1, linetype = "dashed") +
  labs(title = "RCA Over Time - HS09 Sector Products (0901-0910)",
       x = "Year", y = "RCA") +
  theme_bw()
```



10 CEREALS (needs fix, ask Prof. Valerie)

No 4 digit HS classification code is omitted.

```
mx_xports_sector_10 <- read_csv("mx_xports_prod_i/mx_xports_sector_10.csv")
world_xports_sector_10 <- read_csv("world_xports_prod_i/world_xports_sector_10.csv")
mx_xports_sector_10 <- necessary_variables(mx_xports_sector_10)
world_xports_sector_10 <- necessary_variables(world_xports_sector_10)
```

RCA of sector 1001 to 1008

```
# inputs need: country_sector_df, world_sector_df, country_all_df, world_all_df

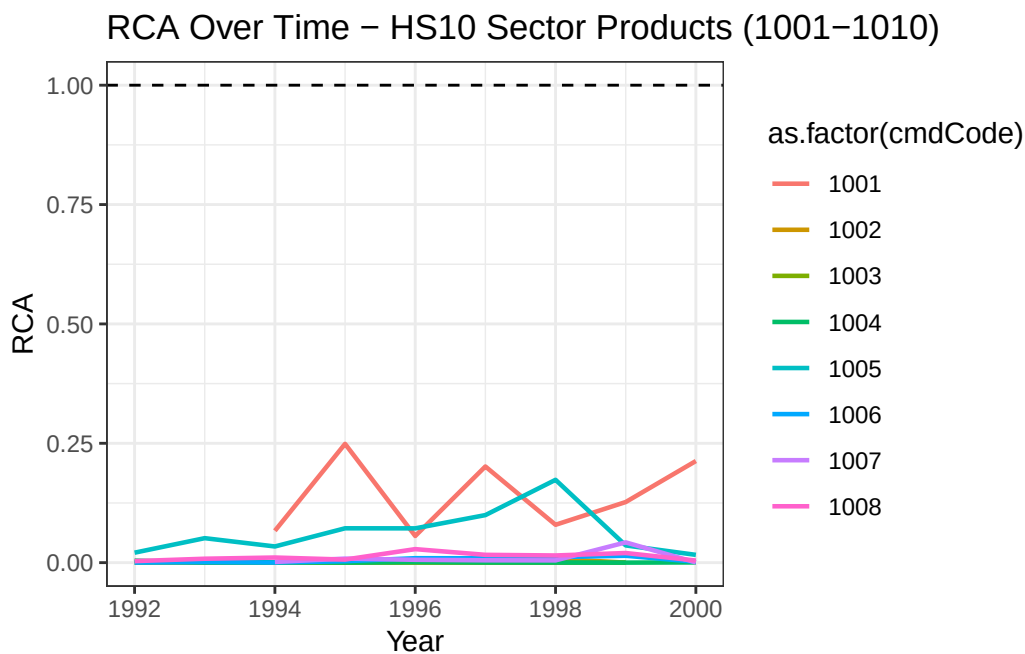
# have: mx_xports_sector_10, world_xports_sector_10, mx_xports_all, world_xports_all

rca_sector_10 <- calculate_rca_generalized(mx_xports_sector_10, world_xports_sector_10, mx_xp
```

Plot sector 10

Visualize over time:

```
## ggplot to bisualize over time
ggplot(rca_sector_10, aes(x = period, y = RCA, color = as.factor(cmdCode), group = cmdCode))
  geom_line(size = 0.8) +
  geom_hline(yintercept = 1, linetype = "dashed") +
  labs(title = "RCA Over Time - HS10 Sector Products (1001-1010)",
       x = "Year", y = "RCA") +
  theme_bw()
```



11 Milling products; malt; starch, etc.; gluten

No 4 digit HS classification code is omitted.

```
mx_xports_sector_11 <- read_csv("mx_xports_prod_i/mx_xports_sector_11.csv")

world_xports_sector_11 <- read_csv("world_xports_prod_i/world_xports_sector_11.csv")

mx_xports_sector_11 <- necessary_variables(mx_xports_sector_11)
world_xports_sector_11 <- necessary_variables(world_xports_sector_11)
```

RCA of sector 1101 to 1109

```
# inputs need: country_sector_df, world_sector_df, country_all_df, world_all_df

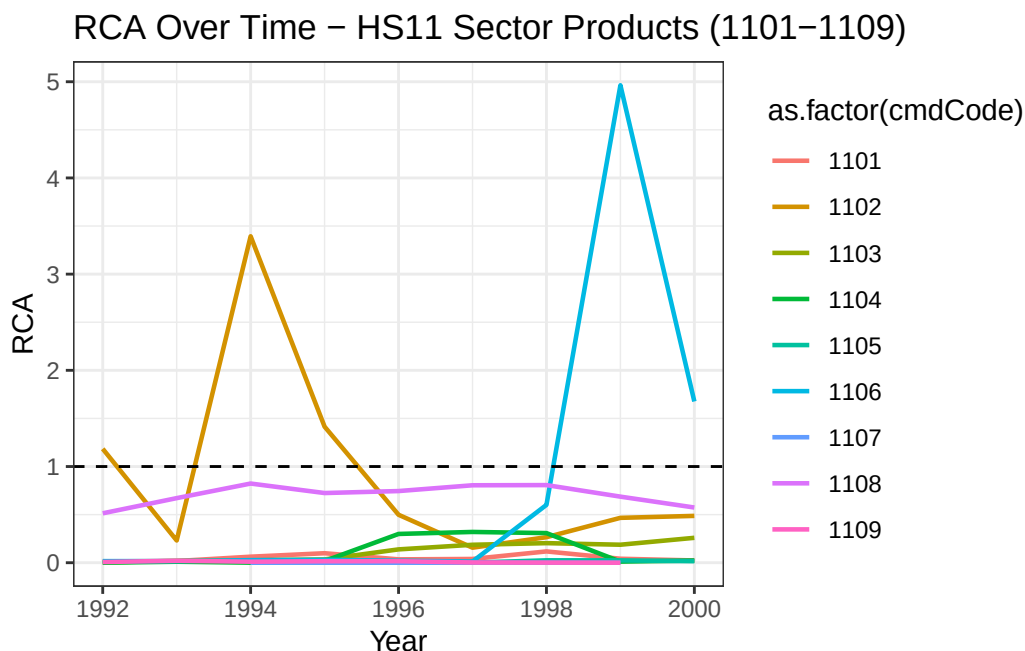
# have: mx_xports_sector_10, world_xports_sector_10, mx_xports_all, world_xports_all

rca_sector_11 <- calculate_rca_generalized(mx_xports_sector_11, world_xports_sector_11, mx_xp
```

Plot sector 11

Visualize over time:

```
## ggplot to bisualize over time
ggplot(rca_sector_11, aes(x = period, y = RCA, color = as.factor(cmdCode), group = cmdCode))
  geom_line(size = 0.8) +
  geom_hline(yintercept = 1, linetype = "dashed") +
  labs(title = "RCA Over Time - HS11 Sector Products (1101-1109)",
        x = "Year", y = "RCA") +
  theme_bw()
```



12 Oilseeds; kernels; industrial medicinal plants; feed

We omit the following HS Codes:

- 1205 Rape or colza seeds; whether or not broken
- 1211 Plants and parts of plants (including seeds and fruits), of a kind used primarily in perfumery, in pharmacy or for insecticidal, fungicidal or similar purposes, fresh, chilled, frozen or dried, whether or not cut, crushed or powdered
- 1212 Locust beans, seaweeds and other algae, sugar beet, sugar cane, fresh, chilled, frozen or dried, whether or not ground; fruit stones, kernels and other vegetable products (including unroasted chicory roots) used primarily for human consumption, n.e.c.
- 1214 Swedes, mangolds, fodder roots, hay, lucerne (alfalfa), clover, sainfoin, forage kale, lupines, vetches and similar forage products, whether or not in the form of pellets

No 4 digit HS classification code is omitted.

```
mx_xports_sector_12 <- read_csv("mx_xports_prod_i/mx_xports_sector_12.csv")

world_xports_sector_12 <- read_csv("world_xports_prod_i/world_xports_sector_12.csv")

mx_xports_sector_12 <- necessary_variables(mx_xports_sector_12)
world_xports_sector_12 <- necessary_variables(world_xports_sector_12)
```

```
mx_xports_sector_12 <- mx_xports_sector_12 %>%
  mutate(fobvalue = as.numeric(as.character(fobvalue)))

world_xports_sector_12 <- world_xports_sector_12 %>%
  mutate(fobvalue = as.numeric(as.character(fobvalue)))
```

RCA of sector 12

```
# inputs need: country_sector_df, world_sector_df, country_all_df, world_all_df

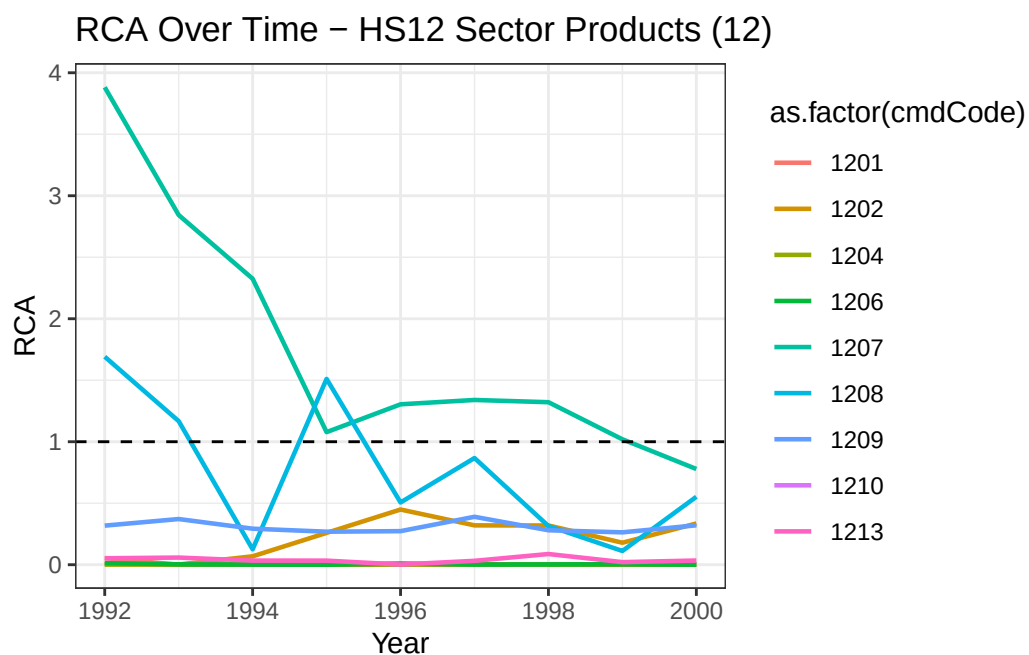
# have: mx_xports_sector_10, world_xports_sector_10, mx_xports_all, world_xports_all

rca_sector_12 <- calculate_rca_generalized(mx_xports_sector_12, world_xports_sector_12, mx_xp
```

Plot sector 12

Visualize over time:

```
## ggplot to bisualize over time
ggplot(rca_sector_12, aes(x = period, y = RCA, color = as.factor(cmdCode), group = cmdCode))
  geom_line(size = 0.8) +
  geom_hline(yintercept = 1, linetype = "dashed") +
  labs(title = "RCA Over Time - HS12 Sector Products (12)",
        x = "Year", y = "RCA") +
  theme_bw()
```



13 Shellac; gum, fat and other plant liquids, juice

```
mx_xports_sector_13 <- read_csv("mx_xports_prod_i/mx_xports_sector_13.csv")
world_xports_sector_13 <- read_csv("world_xports_prod_i/world_xports_sector_13.csv")
mx_xports_sector_13 <- necessary_variables(mx_xports_sector_13)
world_xports_sector_13 <- necessary_variables(world_xports_sector_13)
```

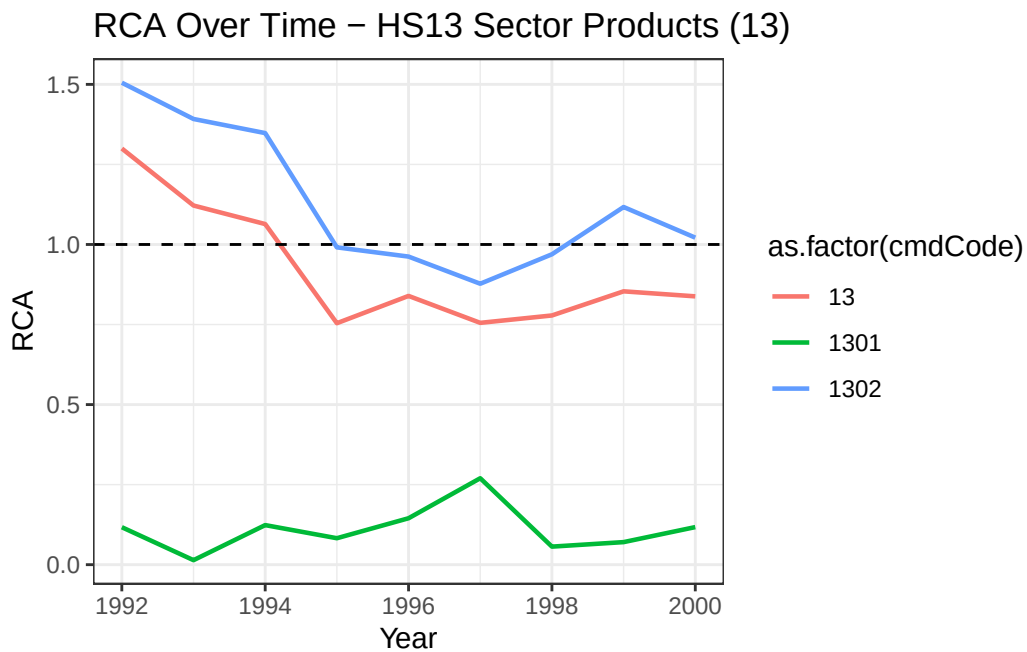
RCA Sector 13

```
# inputs need: country_sector_df, world_sector_df, country_all_df, world_all_df
# have: mx_xports_sector_10, world_xports_sector_10, mx_xports_all, world_xports_all
rca_sector_13 <- calculate_rca_generalized(mx_xports_sector_13, world_xports_sector_13, mx_xp
```

Plot sector 13

Visualize over time:

```
## ggplot to bisualize over time
ggplot(rca_sector_13, aes(x = period, y = RCA, color = as.factor(cmdCode), group = cmdCode))
  geom_line(size = 0.8) +
  geom_hline(yintercept = 1, linetype = "dashed") +
  labs(title = "RCA Over Time - HS13 Sector Products (13)",
       x = "Year", y = "RCA") +
  theme_bw()
```



14 Planting materials for planting; other plant products

```
mx_xports_sector_14 <- read_csv("mx_xports_prod_i/mx_xports_sector_14.csv")
world_xports_sector_14 <- read_csv("world_xports_prod_i/world_xports_sector_14.csv")
mx_xports_sector_14 <- necessary_variables(mx_xports_sector_14)
world_xports_sector_14 <- necessary_variables(world_xports_sector_14)
```

RCA Sector 14

```
# inputs need: country_sector_df, world_sector_df, country_all_df, world_all_df

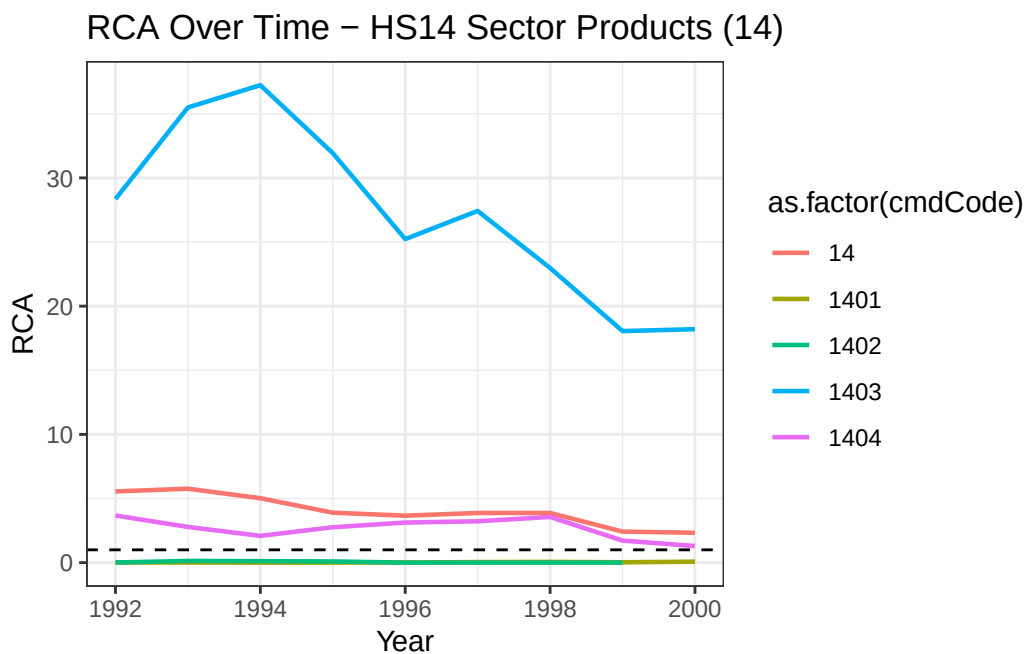
# have: mx_xports_sector_10, world_xports_sector_10, mx_xports_all, world_xports_all

rca_sector_14 <- calculate_rca_generalized(mx_xports_sector_14, world_xports_sector_14, mx_xp
```

Plot sector 14

Visualize over time:

```
## ggplot to bisualize over time
ggplot(rca_sector_14, aes(x = period, y = RCA, color = as.factor(cmdCode), group = cmdCode)) +
  geom_line(size = 0.8) +
  geom_hline(yintercept = 1, linetype = "dashed") +
  labs(title = "RCA Over Time - HS14 Sector Products (14)",
       x = "Year", y = "RCA") +
  theme_bw()
```



15 Animal and vegetable oils and fats; refined edible oils and fats

We Omit the following:

- 1503

- 1505
- 1506
- -1508
- 1510
- 1511 to 1514
- 1517 to 1519

```
mx_xports_sector_15 <- read_csv("mx_xports_prod_i/mx_xports_sector_15.csv")
world_xports_sector_15 <- read_csv("world_xports_prod_i/world_xports_sector_15.csv")

mx_xports_sector_15 <- necessary_variables(mx_xports_sector_15)
world_xports_sector_15 <- necessary_variables(world_xports_sector_15)
```

RCA Sector 15

```
# inputs need: country_sector_df, world_sector_df, country_all_df, world_all_df

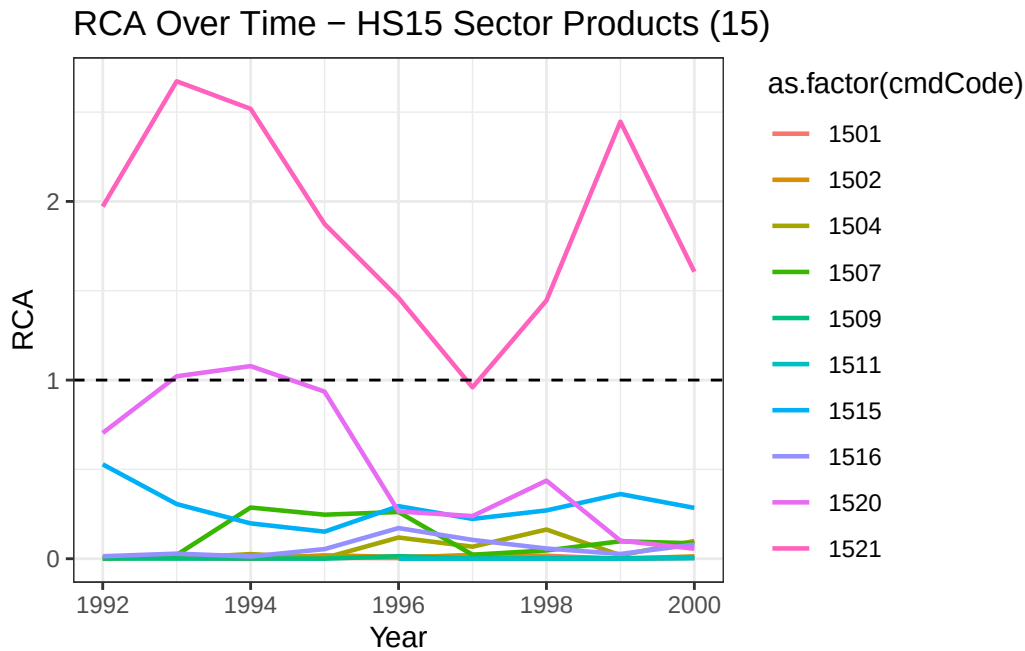
# have: mx_xports_sector_10, world_xports_sector_10, mx_xports_all, world_xports_all

rca_sector_15 <- calculate_rca_generalized(mx_xports_sector_15, world_xports_sector_15, mx_xp
```

Plot sector 15

Visualize over time:

```
## ggplot to bisualize over time
ggplot(rca_sector_15, aes(x = period, y = RCA, color = as.factor(cmdCode), group = cmdCode))
  geom_line(size = 0.8) +
  geom_hline(yintercept = 1, linetype = "dashed") +
  labs(title = "RCA Over Time - HS15 Sector Products (15)",
        x = "Year", y = "RCA") +
  theme_bw()
```



16 Meat, fish and other aquatic invertebrate products

RCA Sector 16

```
mx_xports_sector_16 <- read_csv("mx_xports_prod_i/mx_xports_sector_16.csv")
world_xports_sector_16 <- read_csv("world_xports_prod_i/world_xports_sector_16.csv")

mx_xports_sector_16 <- necessary_variables(mx_xports_sector_16)
world_xports_sector_16 <- necessary_variables(world_xports_sector_16)

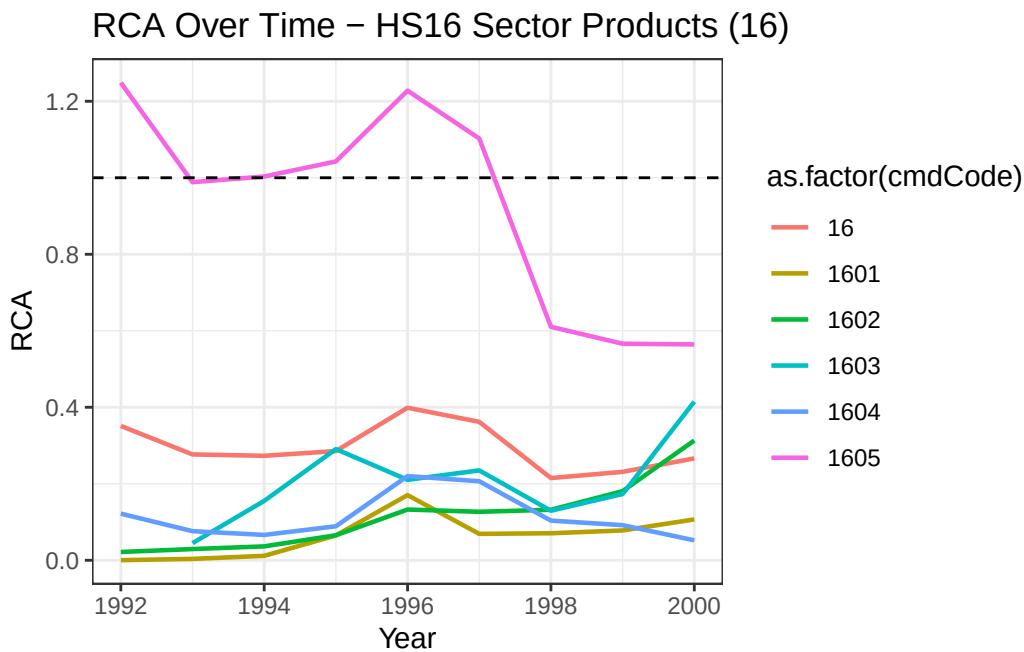
# inputs need: country_sector_df, world_sector_df, country_all_df, world_all_df
# have: mx_xports_sector_10, world_xports_sector_10, mx_xports_all, world_xports_all

rca_sector_16 <- calculate_rca_generalized(mx_xports_sector_16, world_xports_sector_16, mx_xp
```

Plot sector 16

Visualize over time:

```
## ggplot to bisualize over time
ggplot(rca_sector_16, aes(x = period, y = RCA, color = as.factor(cmdCode), group = cmdCode))
  geom_line(size = 0.8) +
  geom_hline(yintercept = 1, linetype = "dashed") +
  labs(title = "RCA Over Time - HS16 Sector Products (16)",
       x = "Year", y = "RCA") +
  theme_bw()
```



17 Sugar and confectionery

RCA Sector 17

```
mx_xports_sector_17 <- read_csv("mx_xports_prod_i/mx_xports_sector_17.csv")
world_xports_sector_17<- read_csv("world_xports_prod_i/world_xports_sector_17.csv")
mx_xports_sector_17 <- necessary_variables(mx_xports_sector_17)
world_xports_sector_17 <- necessary_variables(world_xports_sector_17)

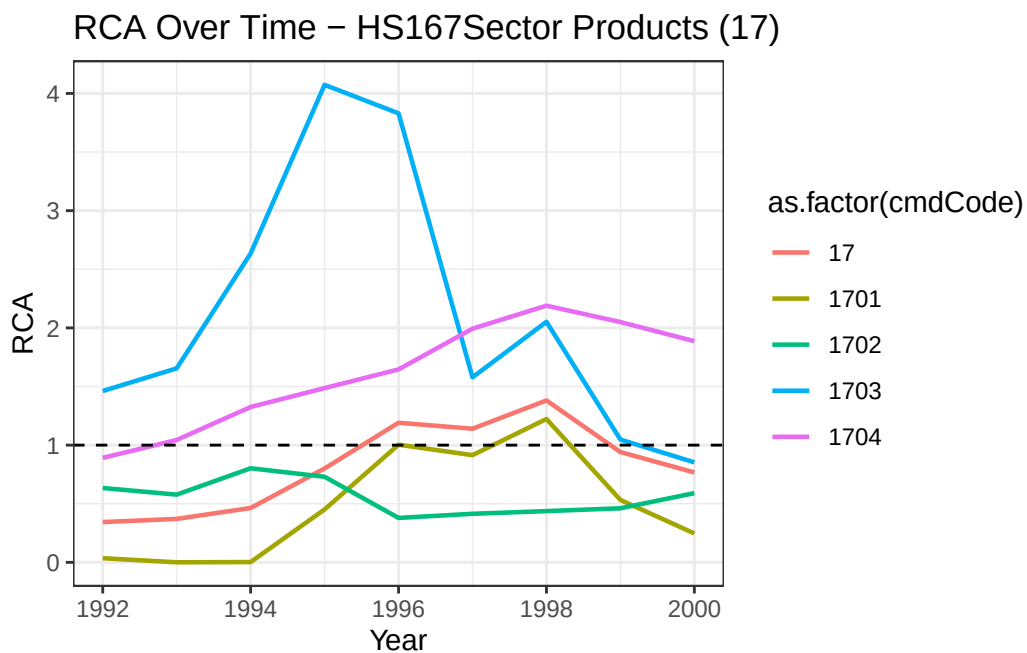
# inputs need: country_sector_df, world_sector_df, country_all_df, world_all_df
# have: mx_xports_sector_10, world_xports_sector_10, mx_xports_all, world_xports_all
```

```
rca_sector_17 <- calculate_rca_generalized(mx_xports_sector_17, world_xports_sector_17, mx_xp
```

Plot sector 16

Visualize over time:

```
## ggplot to bisualize over time
ggplot(rca_sector_17, aes(x = period, y = RCA, color = as.factor(cmdCode), group = cmdCode))
  geom_line(size = 0.8) +
  geom_hline(yintercept = 1, linetype = "dashed") +
  labs(title = "RCA Over Time - HS167Sector Products (17)",
       x = "Year", y = "RCA") +
  theme_bw()
```



18 Cocoa and cocoa products

RCA Sector 18

```
mx_xports_sector_18 <- read_csv("mx_xports_prod_i/mx_xports_sector_18.csv")
world_xports_sector_18 <- read_csv("world_xports_prod_i/world_xports_sector_18.csv")
```

```

mx_xports_sector_18 <- necessary_variables(mx_xports_sector_18)
world_xports_sector_18 <- necessary_variables(world_xports_sector_18)

# inputs need: country_sector_df, world_sector_df, country_all_df, world_all_df

# have: mx_xports_sector_10, world_xports_sector_10, mx_xports_all, world_xports_all

rca_sector_18 <- calculate_rca_generalized(mx_xports_sector_18, world_xports_sector_18, mx_xp

```

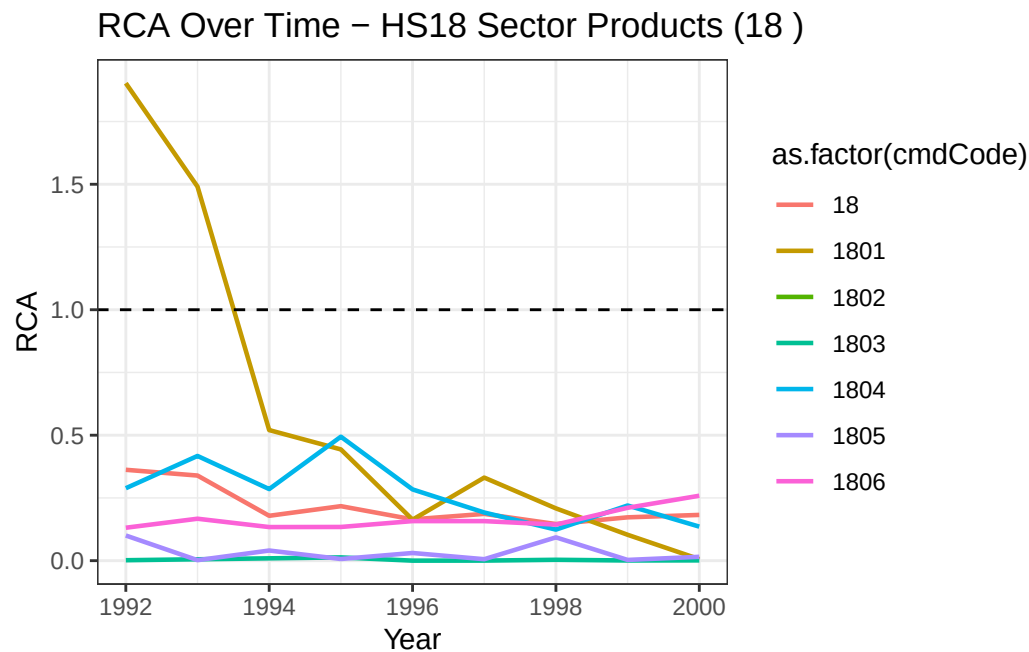
Plot sector 18

Visualize over time:

```

## ggplot to bisualize over time
ggplot(rca_sector_18, aes(x = period, y = RCA, color = as.factor(cmdCode), group = cmdCode))
  geom_line(size = 0.8) +
  geom_hline(yintercept = 1, linetype = "dashed") +
  labs(title = "RCA Over Time - HS18 Sector Products (18 )",
       x = "Year", y = "RCA") +
  theme_bw()

```



19 Cereal flour, starch, etc. or dairy products; cakes

RCA Sector 19

No HS Code is omitted

```
mx_xports_sector_19 <- read_csv("mx_xports_prod_i/mx_xports_sector_19.csv")

world_xports_sector_19 <- read_csv("world_xports_prod_i/world_xports_sector_19.csv")

mx_xports_sector_19 <- necessary_variables(mx_xports_sector_19)
world_xports_sector_19 <- necessary_variables(world_xports_sector_19)

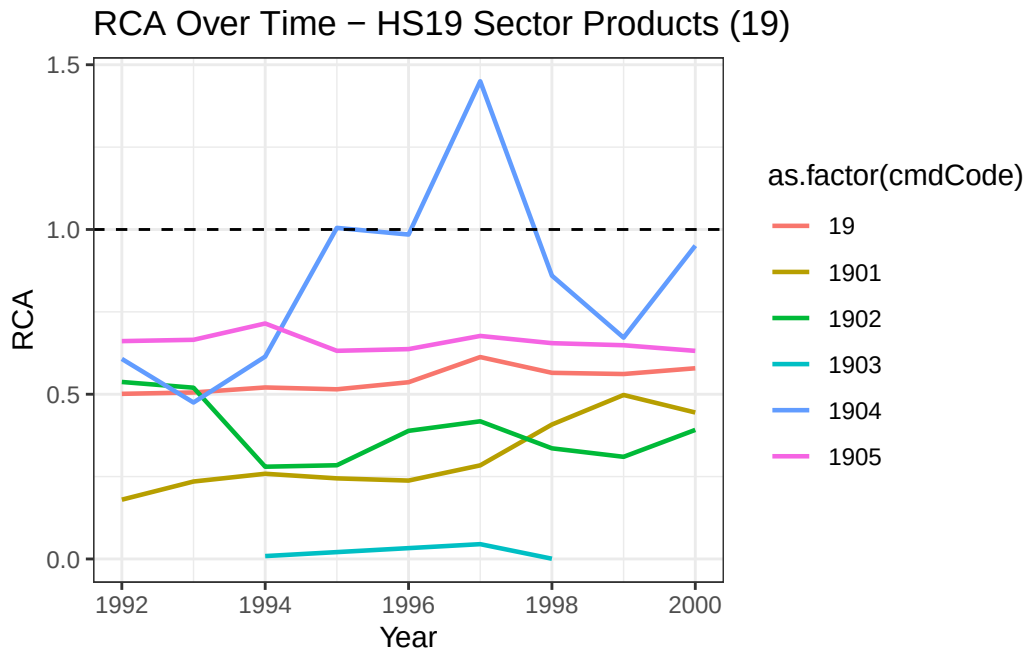
# inputs need: country_sector_df, world_sector_df, country_all_df, world_all_df
# have: mx_xports_sector_10, world_xports_sector_10, mx_xports_all, world_xports_all

rca_sector_19 <- calculate_rca_generalized(mx_xports_sector_19, world_xports_sector_19, mx_xp
```

Plot sector 19

Visualize over time:

```
## ggplot to bisualize over time
ggplot(rca_sector_19, aes(x = period, y = RCA, color = as.factor(cmdCode), group = cmdCode))
  geom_line(size = 0.8) +
  geom_hline(yintercept = 1, linetype = "dashed") +
  labs(title = "RCA Over Time - HS19 Sector Products (19)",
        x = "Year", y = "RCA") +
  theme_bw()
```



20 Products of vegetables, fruits or other parts of plants

RCA Sector 20

No HS Code is omitted

```
mx_xports_sector_20 <- read_csv("mx_xports_prod_i/mx_xports_sector_20.csv")

world_xports_sector_20 <- read_csv("world_xports_prod_i/world_xports_sector_20.csv")

mx_xports_sector_20 <- necessary_variables(mx_xports_sector_20)
world_xports_sector_20 <- necessary_variables(world_xports_sector_20)

# inputs need: country_sector_df, world_sector_df, country_all_df, world_all_df

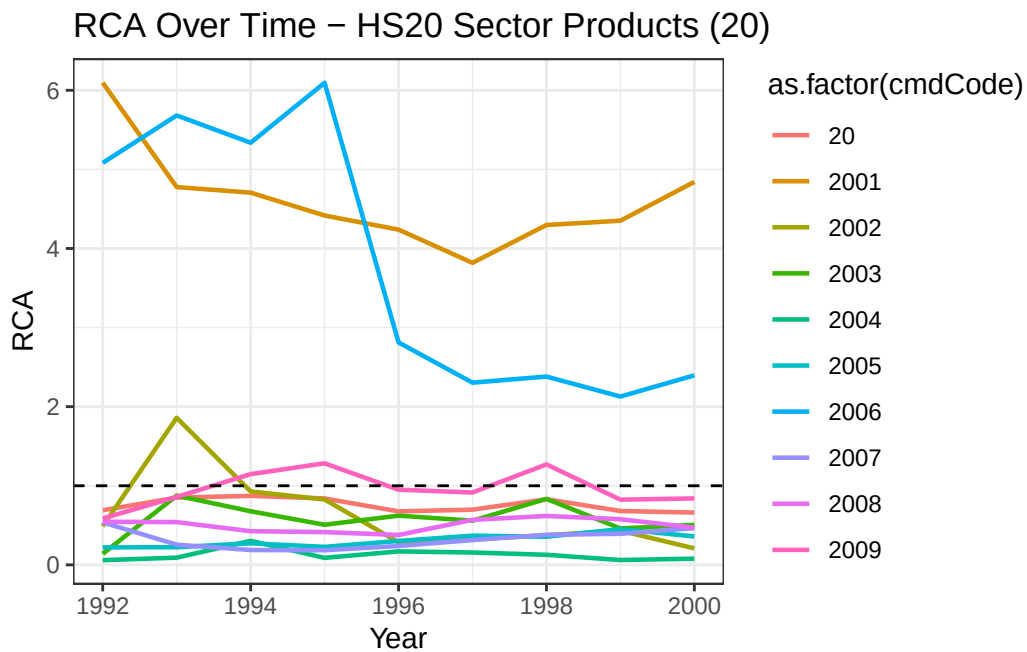
# have: mx_xports_sector_10, world_xports_sector_10, mx_xports_all, world_xports_all

rca_sector_20 <- calculate_rca_generalized(mx_xports_sector_20, world_xports_sector_20, mx_xp
```

Plot sector 20

Visualize over time:

```
## ggplot to bisualize over time
ggplot(rca_sector_20, aes(x = period, y = RCA, color = as.factor(cmdCode), group = cmdCode))
  geom_line(size = 0.8) +
  geom_hline(yintercept = 1, linetype = "dashed") +
  labs(title = "RCA Over Time - HS20 Sector Products (20)",
       x = "Year", y = "RCA") +
  theme_bw()
```



21 Miscellaneous food

RCA Sector 21

No HS Code is omitted

```
mx_xports_sector_21 <- read_csv("mx_xports_prod_i/mx_xports_sector_21.csv")

world_xports_sector_21 <- read_csv("world_xports_prod_i/world_xports_sector_21.csv")

mx_xports_sector_21 <- necessary_variables(mx_xports_sector_21)
world_xports_sector_21 <- necessary_variables(world_xports_sector_21)
```

```
# inputs need: country_sector_df, world_sector_df, country_all_df, world_all_df

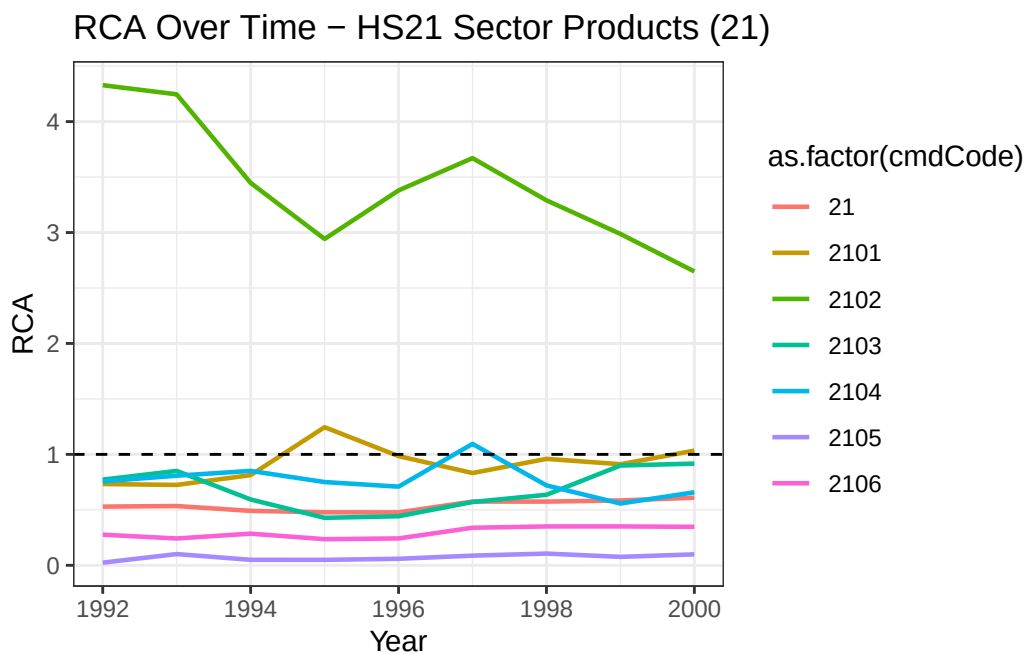
# have: mx_xports_sector_10, world_xports_sector_10, mx_xports_all, world_xports_all

rca_sector_21 <- calculate_rca_generalized(mx_xports_sector_21, world_xports_sector_21, mx_xp
```

Plot sector 21

Visualize over time:

```
## ggplot to bisualize over time
ggplot(rca_sector_21, aes(x = period, y = RCA, color = as.factor(cmdCode), group = cmdCode))
  geom_line(size = 0.8) +
  geom_hline(yintercept = 1, linetype = "dashed") +
  labs(title = "RCA Over Time - HS21 Sector Products (21)",
       x = "Year", y = "RCA") +
  theme_bw()
```



22 Beverages, wine and vinegar

RCA Sector 22

No HS Code is omitted

```

mx_xports_sector_22 <- read_csv("mx_xports_prod_i/mx_xports_sector_22.csv")

world_xports_sector_22 <- read_csv("world_xports_prod_i/world_xports_sector_22.csv")

mx_xports_sector_22 <- necessary_variables(mx_xports_sector_22)
world_xports_sector_22 <- necessary_variables(world_xports_sector_22)

# inputs need: country_sector_df, world_sector_df, country_all_df, world_all_df

# have: mx_xports_sector_10, world_xports_sector_10, mx_xports_all, world_xports_all

rca_sector_22 <- calculate_rca_generalized(mx_xports_sector_22, world_xports_sector_22, mx_xp

```

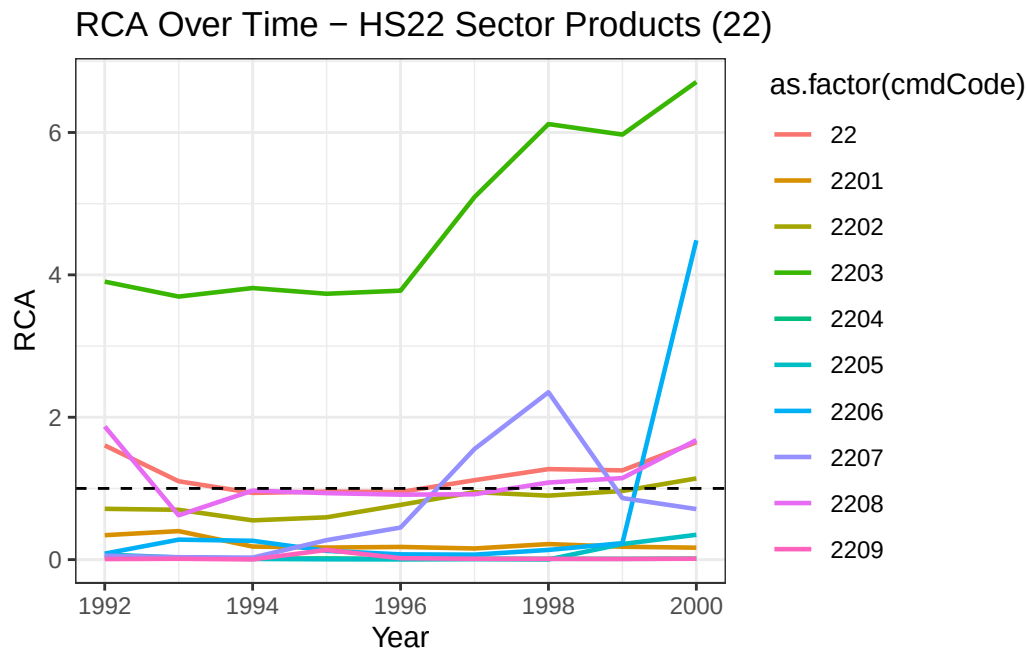
Plot sector 22

Visualize over time:

```

## ggplot to bisualize over time
ggplot(rca_sector_22, aes(x = period, y = RCA, color = as.factor(cmdCode), group = cmdCode)) +
  geom_line(size = 0.8) +
  geom_hline(yintercept = 1, linetype = "dashed") +
  labs(title = "RCA Over Time - HS22 Sector Products (22)",
       x = "Year", y = "RCA") +
  theme_bw()

```



23 Food industry residues and waste; formulated feed

RCA Sector 23

No HS Code is omitted

```
mx_xports_sector_23 <- read_csv("mx_xports_prod_i/mx_xports_sector_23.csv")

world_xports_sector_23 <- read_csv("world_xports_prod_i/world_xports_sector_23.csv")

mx_xports_sector_23 <- necessary_variables(mx_xports_sector_23)
world_xports_sector_23 <- necessary_variables(world_xports_sector_23)

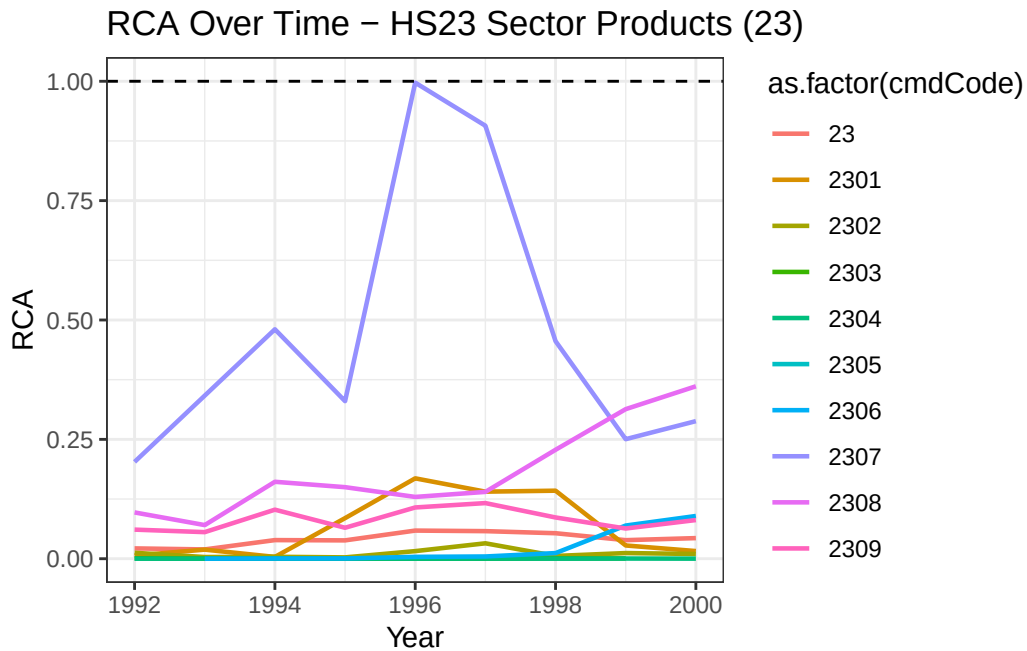
# inputs need: country_sector_df, world_sector_df, country_all_df, world_all_df
# have: mx_xports_sector_10, world_xports_sector_10, mx_xports_all, world_xports_all

rca_sector_23 <- calculate_rca_generalized(mx_xports_sector_23, world_xports_sector_23, mx_xp
```

Plot sector 23

Visualize over time:

```
## ggplot to bisualize over time
ggplot(rca_sector_23, aes(x = period, y = RCA, color = as.factor(cmdCode), group = cmdCode))
  geom_line(size = 0.8) +
  geom_hline(yintercept = 1, linetype = "dashed") +
  labs(title = "RCA Over Time - HS23 Sector Products (23)",
        x = "Year", y = "RCA") +
  theme_bw()
```



24 Tobacco, tobacco and tobacco substitute products

RCA Sector 24

No HS Code is omitted.

```
mx_xports_sector_24 <- read_csv("mx_xports_prod_i/mx_xports_sector_24.csv")
world_xports_sector_24 <- read_csv("world_xports_prod_i/world_xports_sector_24.csv")
mx_xports_sector_24 <- necessary_variables(mx_xports_sector_24)
world_xports_sector_24 <- necessary_variables(world_xports_sector_24)
```

```
# inputs need: country_sector_df, world_sector_df, country_all_df, world_all_df
```

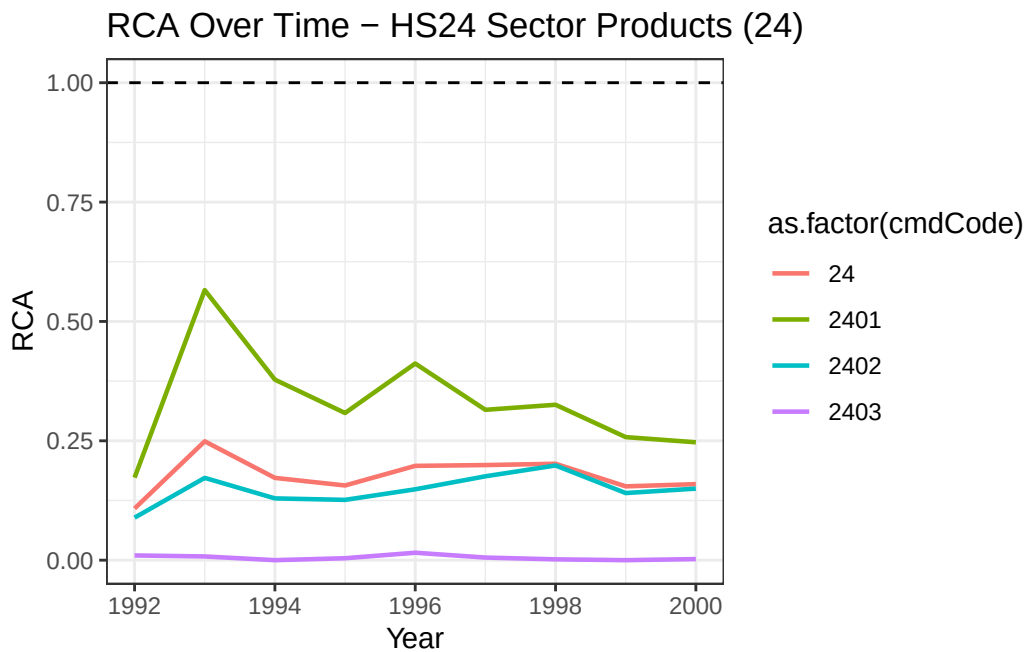
```
# have: mx_xports_sector_10, world_xports_sector_10, mx_xports_all, world_xports_all
```

```
rca_sector_24 <- calculate_rca_generalized(mx_xports_sector_24, world_xports_sector_24, mx_xp
```

Plot sector 24

Visualize over time:

```
## ggplot to bisualize over time
ggplot(rca_sector_24, aes(x = period, y = RCA, color = as.factor(cmdCode), group = cmdCode))
  geom_line(size = 0.8) +
  geom_hline(yintercept = 1, linetype = "dashed") +
  labs(title = "RCA Over Time - HS24 Sector Products (24)",
       x = "Year", y = "RCA") +
  theme_bw()
```



50 Silk

RCA Sector 50

No HS Code is omitted

```
mx_xports_sector_50 <- read_csv("mx_xports_prod_i/mx_xports_sector_50.csv")
world_xports_sector_50 <- read_csv("world_xports_prod_i/world_xports_sector_50.csv")
mx_xports_sector_50 <- necessary_variables(mx_xports_sector_50)
world_xports_sector_50 <- necessary_variables(world_xports_sector_50)
```

```
# inputs need: country_sector_df, world_sector_df, country_all_df, world_all_df

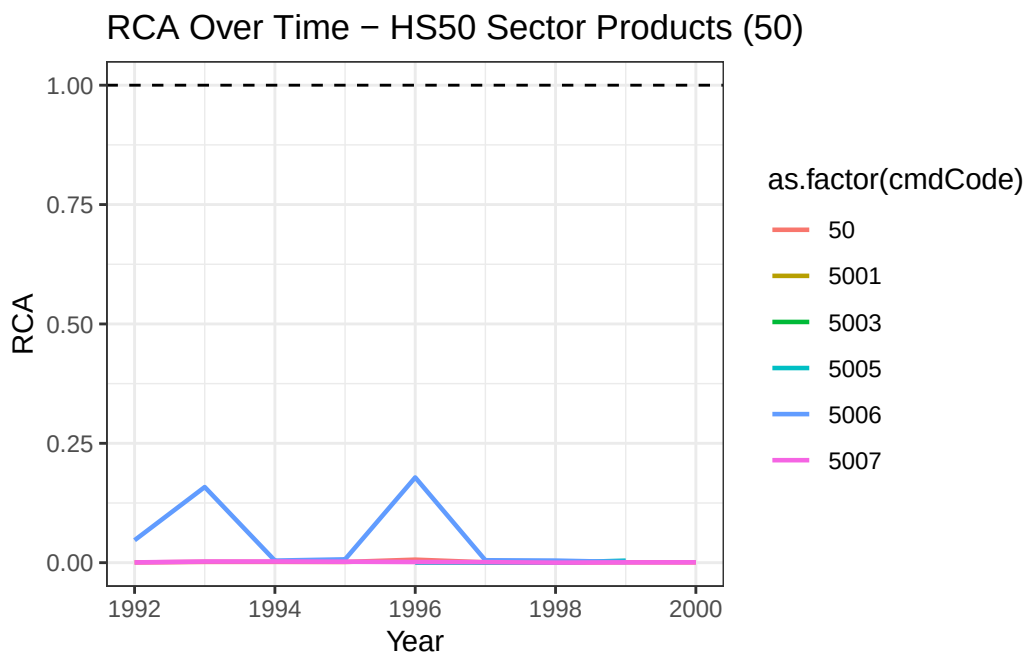
# have: mx_xports_sector_10, world_xports_sector_10, mx_xports_all, world_xports_all

rca_sector_50 <- calculate_rca_generalized(mx_xports_sector_50, world_xports_sector_50, mx_xp
```

Plot sector 23

Visualize over time:

```
## ggplot to bisualize over time
ggplot(rca_sector_50, aes(x = period, y = RCA, color = as.factor(cmdCode), group = cmdCode)) +
  geom_line(size = 0.8) +
  geom_hline(yintercept = 1, linetype = "dashed") +
  labs(title = "RCA Over Time - HS50 Sector Products (50)",
       x = "Year", y = "RCA") +
  theme_bw()
```



51 Wool and other animal hair

RCA Sector 51

We omit the following HS Codes

- 5103 Waste of wool or of fine or coarse animal hair, including yarn waste but excluding garnetted stock
- 5104 Wool, or fine or coarse animal hair; garnetted stock
- 5113 Woven fabrics of coarse animal hair or of horsehair

```
mx_xports_sector_51 <- read_csv("mx_xports_prod_i/mx_xports_sector_51.csv")
world_xports_sector_51 <- read_csv("world_xports_prod_i/world_xports_sector_51.csv")

mx_xports_sector_51 <- necessary_variables(mx_xports_sector_51)
world_xports_sector_51 <- necessary_variables(world_xports_sector_51)

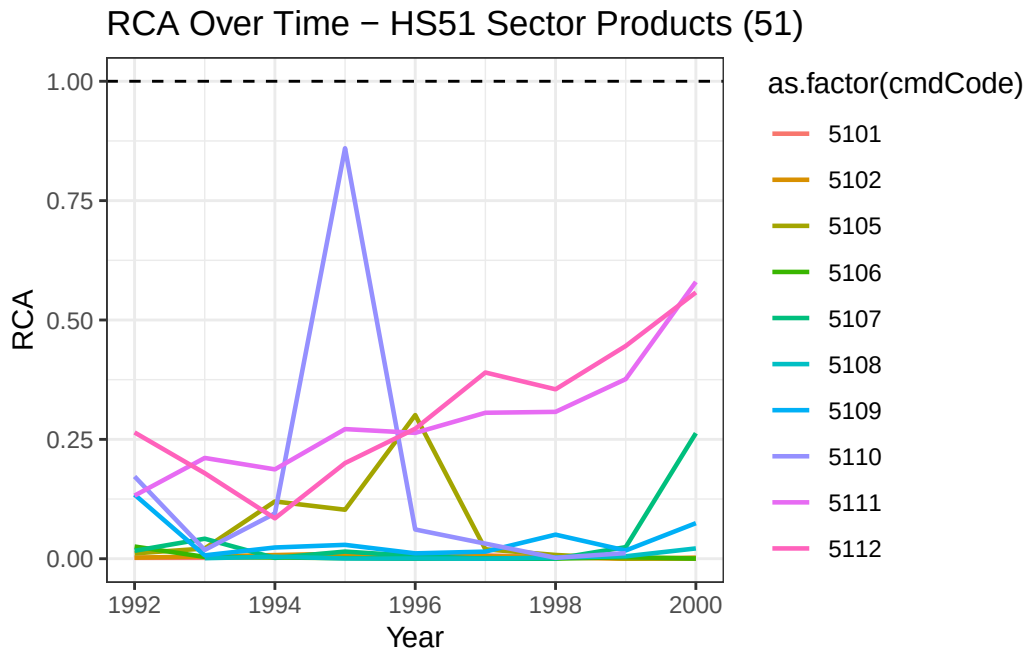
# inputs need: country_sector_df, world_sector_df, country_all_df, world_all_df
# have: mx_xports_sector_10, world_xports_sector_10, mx_xports_all, world_xports_all

rca_sector_51 <- calculate_rca_generalized(mx_xports_sector_51, world_xports_sector_51, mx_xp
```

Plot sector 51

Visualize over time:

```
## ggplot to bisualize over time
ggplot(rca_sector_51, aes(x = period, y = RCA, color = as.factor(cmdCode), group = cmdCode))
  geom_line(size = 0.8) +
  geom_hline(yintercept = 1, linetype = "dashed") +
  labs(title = "RCA Over Time - HS51 Sector Products (51)",
       x = "Year", y = "RCA") +
  theme_bw()
```



52 Cotton

Omit

- 5205 Cotton yarn (other than sewing thread), containing 85% or more by weight of cotton, not put up for retail sale
- 5206 Cotton yarn (other than sewing thread), containing less than 85% by weight of cotton, not put up for retail sale

```
mx_xports_sector_52 <- read_csv("mx_xports_prod_i/mx_xports_sector_52.csv")
world_xports_sector_52 <- read_csv("world_xports_prod_i/world_xports_sector_52.csv")

mx_xports_sector_52 <- necessary_variables(mx_xports_sector_52)
world_xports_sector_52 <- necessary_variables(world_xports_sector_52)

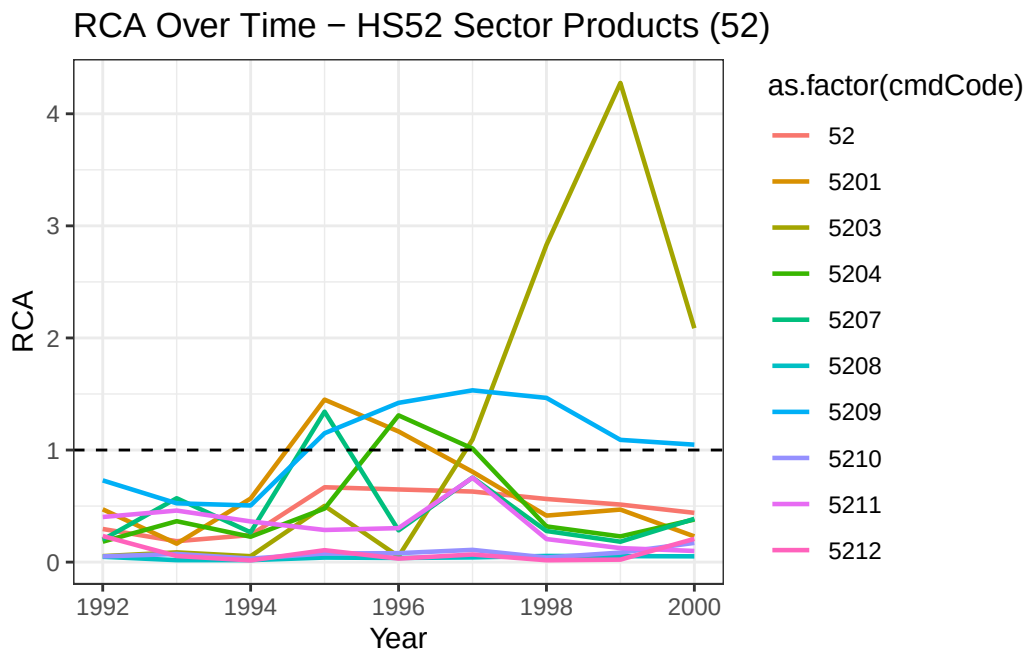
# inputs need: country_sector_df, world_sector_df, country_all_df, world_all_df
# have: mx_xports_sector_10, world_xports_sector_10, mx_xports_all, world_xports_all

rca_sector_52 <- calculate_rca_generalized(mx_xports_sector_52, world_xports_sector_52, mx_xp
```

Plot sector 52

Visualize over time:

```
## ggplot to bisualize over time
ggplot(rca_sector_52, aes(x = period, y = RCA, color = as.factor(cmdCode), group = cmdCode))
  geom_line(size = 0.8) +
  geom_hline(yintercept = 1, linetype = "dashed") +
  labs(title = "RCA Over Time - HS52 Sector Products (52)",
       x = "Year", y = "RCA") +
  theme_bw()
```



53 Other plant fibers

We omit the following:

- 5310 Woven fabrics of jute, other textile bast fibres of heading no. 5303
- 5311 Woven fabrics of other vegetable textile fibres; woven fabrics of paper yarn

```
mx_xports_sector_53<- read_csv("mx_xports_prod_i/mx_xports_sector_53.csv")
world_xports_sector_53 <- read_csv("world_xports_prod_i/world_xports_sector_53.csv")
```

```

mx_xports_sector_53 <- necessary_variables(mx_xports_sector_53)
world_xports_sector_53 <- necessary_variables(world_xports_sector_53)

# inputs need: country_sector_df, world_sector_df, country_all_df, world_all_df

# have: mx_xports_sector_10, world_xports_sector_10, mx_xports_all, world_xports_all

rca_sector_53 <- calculate_rca_generalized(mx_xports_sector_53, world_xports_sector_53, mx_xp

```

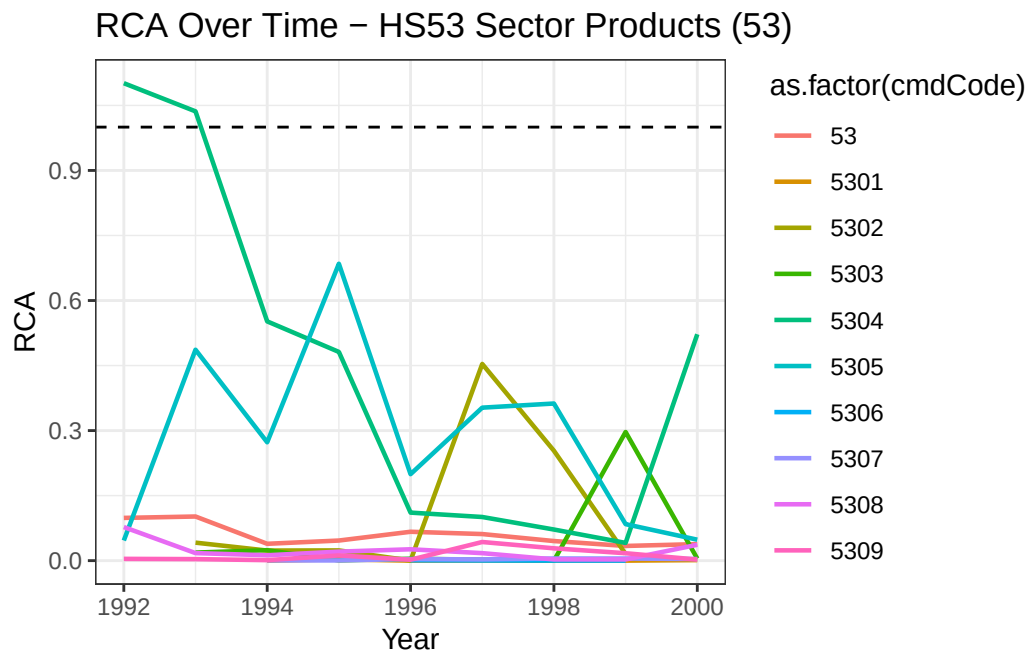
Plot sector 53

Visualize over time:

```

## ggplot to bisualize over time
ggplot(rca_sector_53, aes(x = period, y = RCA, color = as.factor(cmdCode), group = cmdCode))
  geom_line(size = 0.8) +
  geom_hline(yintercept = 1, linetype = "dashed") +
  labs(title = "RCA Over Time - HS53 Sector Products (53)",
       x = "Year", y = "RCA") +
  theme_bw()

```



Miscellaneous and Extra Code

```
##GENERALIZED RCA ANALYSIS

## Generalized RCA function for any product code
#calculate_rca_for_product <- function(product_code) {

  # build file paths for product
  # mx_prod_path <- paste0("mx_xports_prod_i/mx_xports_", product_code, ".csv")
  # world_prod_path <- paste0("world_xports_prod_i/world_xports_", product_code, ".csv")

  # load and read the data
  #mx_xports_prod <- read_csv(mx_prod_path, show_col_types = FALSE)
  # world_xports_prod <- read_csv(world_prod_path, show_col_types = FALSE)

  # load world totals for time frame
  #mx_xports_all <- read_csv("mx_total_xports_world/mx_xports_all.csv", show_col_types = FALSE)
  #world_xports_all <- read_csv("world_total_xports/world_xports_all.csv", show_col_types = FALSE)

  # clean and arrange using necessary_variables function
  # mx_xports_prod <- necessary_variables(mx_xports_prod)
  # world_xports_prod <- necessary_variables(world_xports_prod)
  # mx_xports_all <- necessary_variables(mx_xports_all)
  # world_xports_all <- necessary_variables(world_xports_all)

  # calculate RCA
  # rca_df <- mx_xports_prod %>%
  #   inner_join(mx_xports_all, by = "period", suffix = c("_prod", "_total")) %>%
  #   inner_join(world_xports_prod, by = "period") %>%
  #   inner_join(world_xports_all, by = "period", suffix = c("_world_prod", "_world_total")) %>%
  #   mutate(
  #     RCA = (fobvalue_prod / fobvalue_total) /
  #       (fobvalue_world_prod / fobvalue_world_total)
  #   ) %>%
  #   select(period, RCA) %>%
  #   mutate(product_code = product_code)

  # return(rca_df)
#}
```

run RCA all sectors

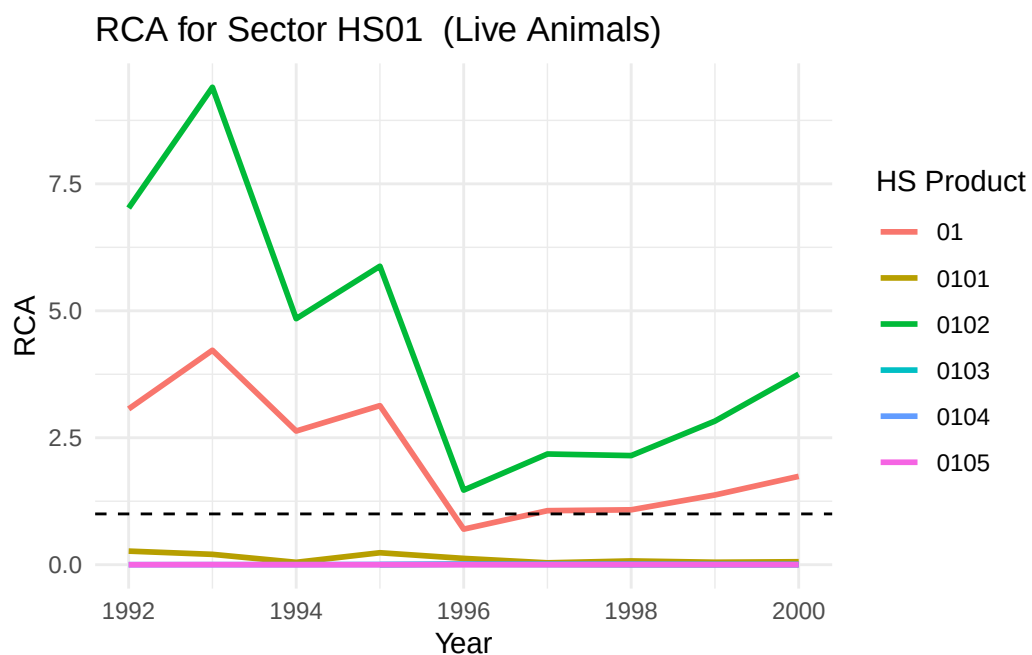
```
rca_all <- purrr::map_dfr(sector_codes, process_sector)

## keep only first two digits

rca_all <- rca_all |>
  mutate(
    cmdCode = as.character(cmdCode),
    sector = substr(cmdCode, 1, 2)
  )
```

plot one sector (look at HS01 Animals: live for reference)

```
ggplot(
  rca_all |> filter(sector == "01"), #filter for first sector
  aes(x = period, y = RCA, color = cmdCode, group = cmdCode)
) +
  geom_line(size = 1) +
  geom_hline(yintercept = 1, linetype = "dashed") +
  labs(
    title = "RCA for Sector HS01 (Live Animals)",
    x = "Year",
    y = "RCA",
    color = "HS Product"
  ) +
  theme_minimal()
```



plot all sectors at once

```
ggplot(rca_all, aes(x = period, y = RCA)) +
  geom_line(color = "steelblue") +
  geom_hline(yintercept = 1, linetype = "dashed") +
  facet_wrap(~cmdCode) +
  theme_minimal() +
  labs(
    title = "RCA Over Time by Product",
    x = "Year",
    y = "RCA"
  )
```

	0502	0503	0504	0505	0506	0507	0508	0510	0701	0702	0703	0704	0705	0706
	0707	0708	0709	0712	0801	0802	0803	0804	0805	0806	0807	0808	0809	0810
	0901	0902	0903	0904	0905	0906	0907	0908	0909	0910	1001	1002	1003	1004
	1005	1006	1007	1008	1101	1102	1103	1104	1105	1106	1107	1108	1109	1201
	1202	1204	1206	1207	1208	1209	1210	1213	13	1301	1302	14	1401	1402
	1403	1404	1501	1502	1504	1507	1509	1511	1515	1516	1520	1521	16	1601
	1602	1603	1604	1605	17	1701	1702	1703	1704	18	1801	1802	1803	1804
	1805	1806	19	1901	1902	1903	1904	1905	20	2001	2002	2003	2004	2005
	2006	2007	2008	2009	21	2101	2102	2103	2104	2105	2106	22	2201	2202
	2203	2204	2205	2206	2207	2208	2209	23	2301	2302	2303	2304	2305	2306
	2307	2308	2309	24	2401	2402	2403	50	5001	5003	5005	5006	5007	5101

```

sectors <- unique(rca_all$sector) ## creatrs a list for all elements in sector_code MAKE SUR

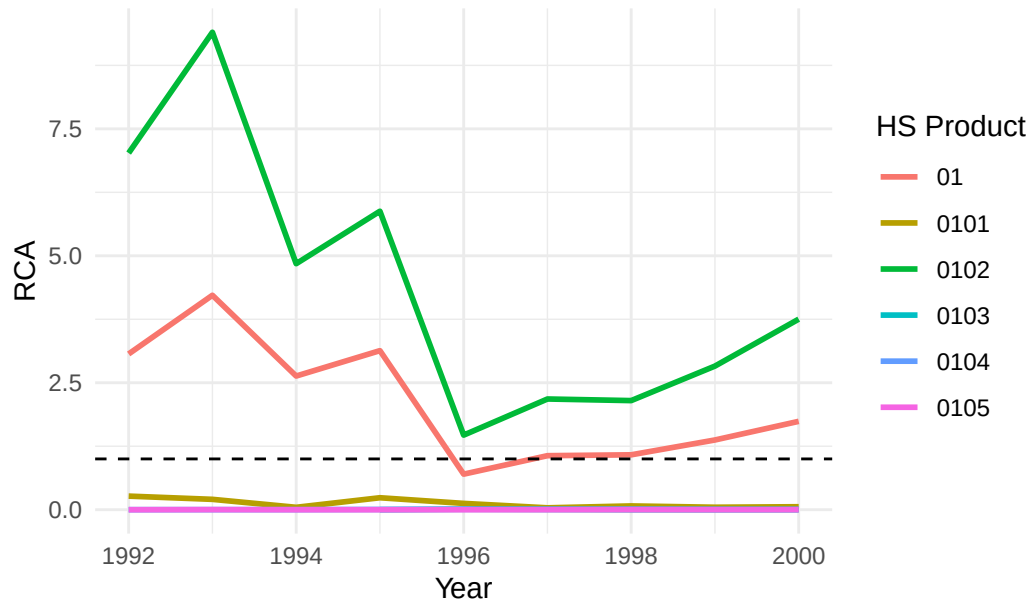
for (s in sectors) { ## for all elements s in list sectors,

  allplots <- ggplot(
    rca_all |> filter(sector == s), ## filter for HS code loops
    aes(x = period, y = RCA, color = cmdCode, group = cmdCode)
  ) +
    geom_line(size = 1) +
    geom_hline(yintercept = 1, linetype = "dashed") +
    labs(
      title = paste("RCA for Sector", s),
      x = "Year",
      y = "RCA",
      color = "HS Product"
    ) +
    theme_minimal()

  print(allplots)
}

```

RCA for Sector 01



RCA for Sector 02

